

Anatomy & Physiology of the Neocortex

NEW YORK INSTITUTE
OF TECHNOLOGY

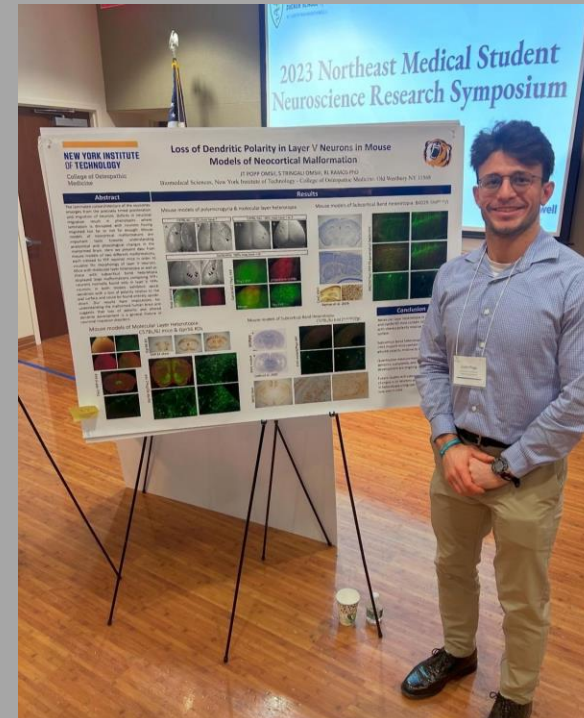
College of Osteopathic
Medicine

PPOM1

- Raddy L. Ramos, PhD
- Assoc. Prof, Biomedical Sciences
- Email: rramos02@nyit.edu



- Justin Thomas Popp
- OMS 4
- Email: Jpopp@nyit.edu

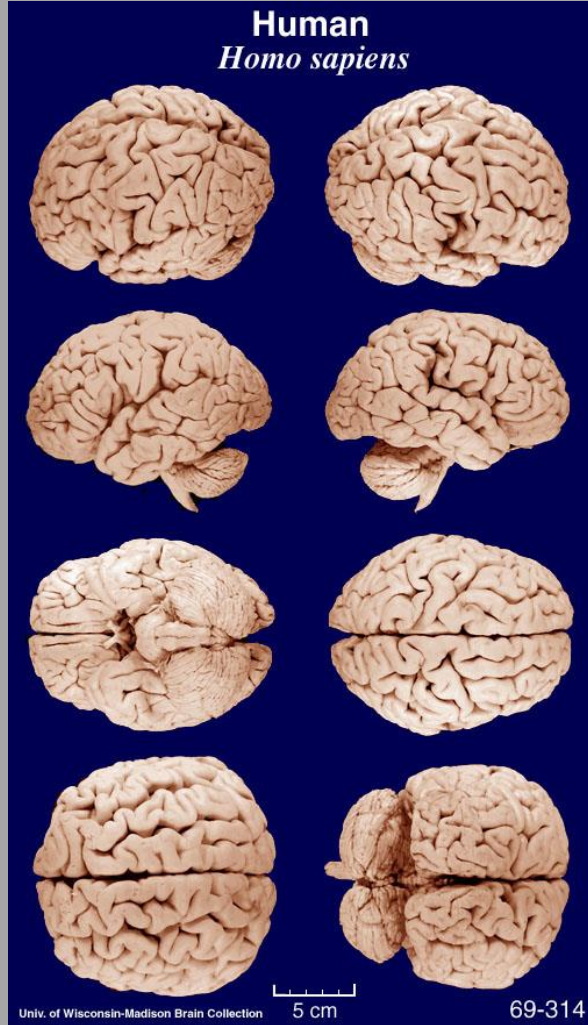


Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session objectives

- Students will be able to describe the gross organization of the neocortex in relation to other structures in the brain as well as organization according to major lobes, fissures, sulci, and gyrii.
- Students will be able to describe the anatomy of neocortical connections including those associated with intracortical, intercortical, and subcortical circuits.
- Students will be able to describe the basic cytoarchitecture of the neocortex as it relates to its six layers.
- Students will be able to describe the anatomy of the major neuron types of the neocortex including the basic physiology of these neurons.
- Students will be able to describe the physiology of neocortical neurons as it relates to sensation, voluntary motor function, attention, and higher mental processes

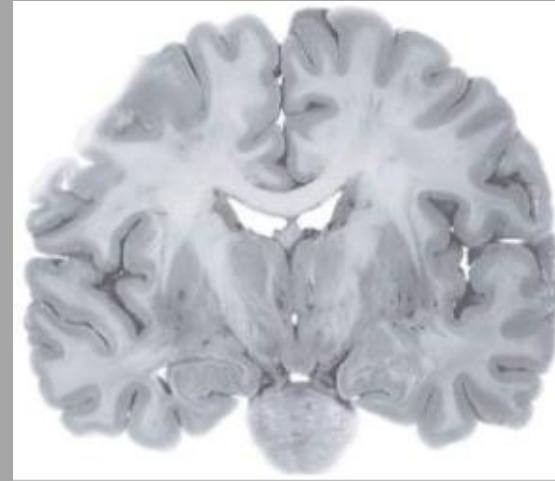
Neocortex: organized by sulci, gyri, & lobes.



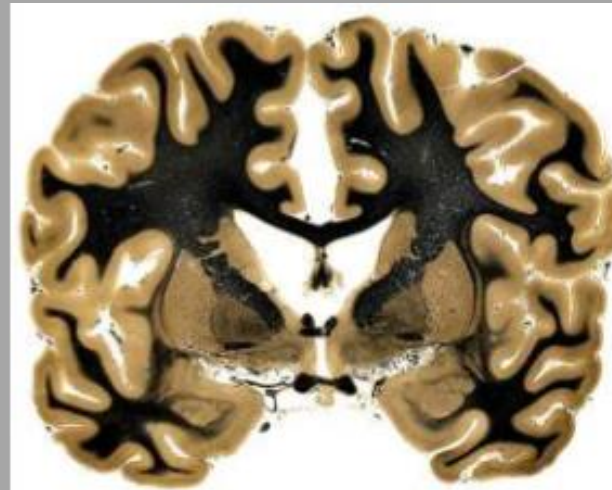
- Gray matter= cell bodies of neurons
- White matter = axons

- It would take over a week to discuss every part of the neocortex. We are covering the major lobes and gyri, involved in sensation, perception behavior etc.

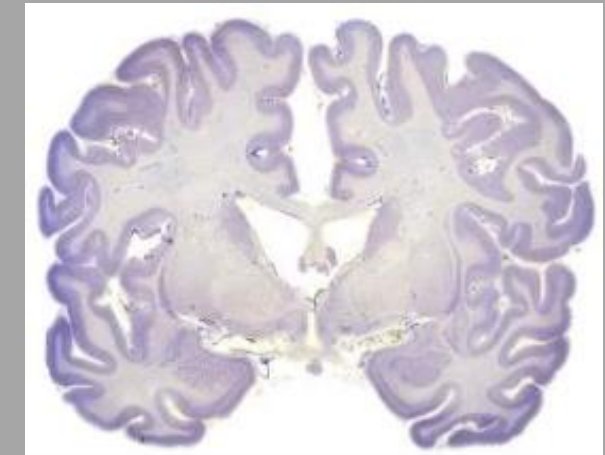
Unstained



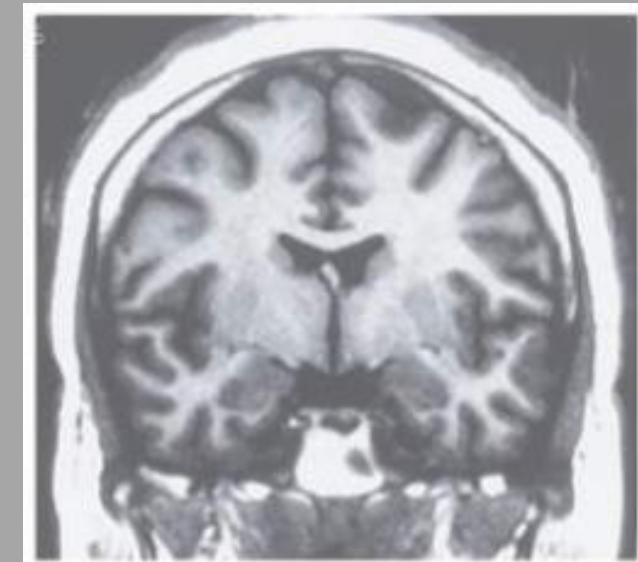
Axonal stain



Cell body stain (Nissl) stains Nissl bodies, which are ER and ribosome of neurons



MRI



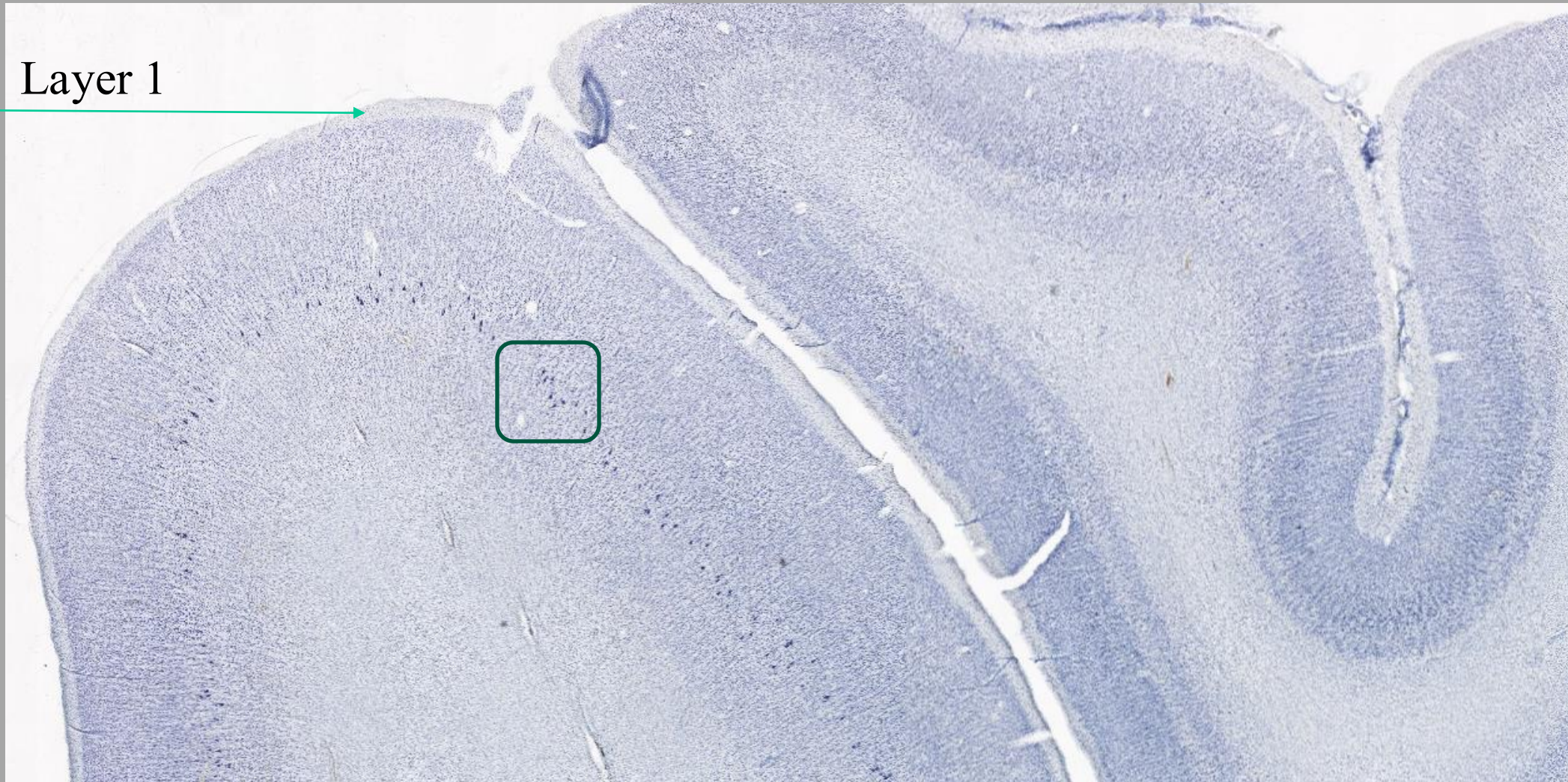
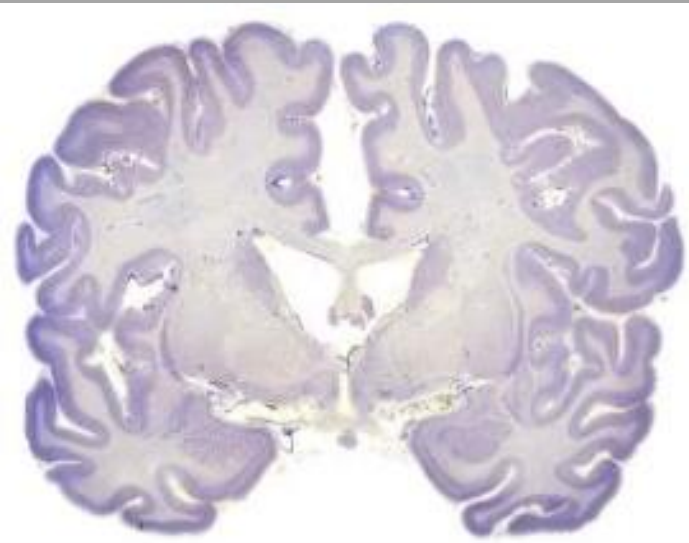
In vivo

Cyto-architecture of the neocortex

- This is nissl stain

What is the distribution, density, and/or pattern of cells in the neocortex?

- Sulci
- Gyri



- Box shows pyramidal cells, see next slide

Neocortex has a 6-layered cytoarchitecture. Layer 1 has very few neurons and mostly axons

Cytology of the neocortex

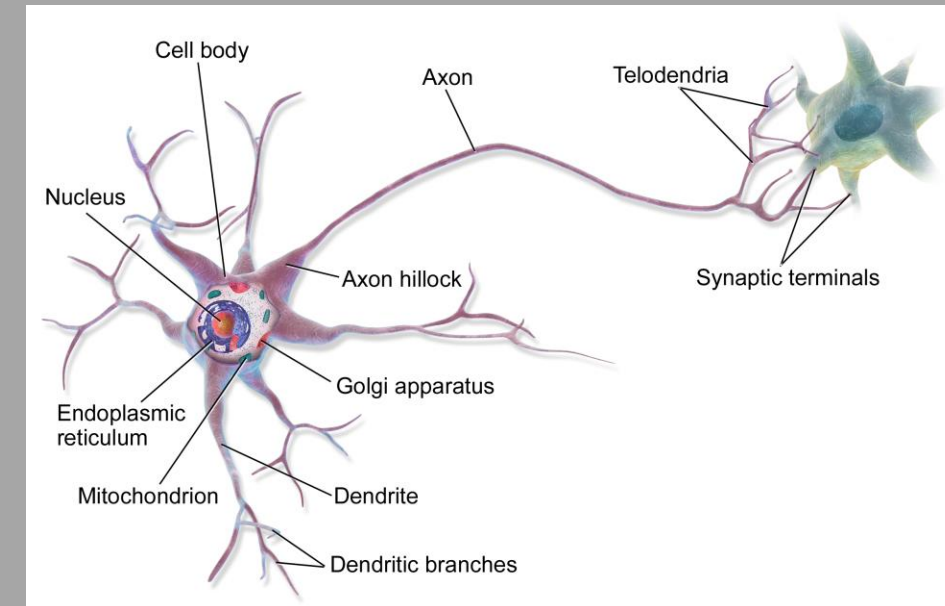
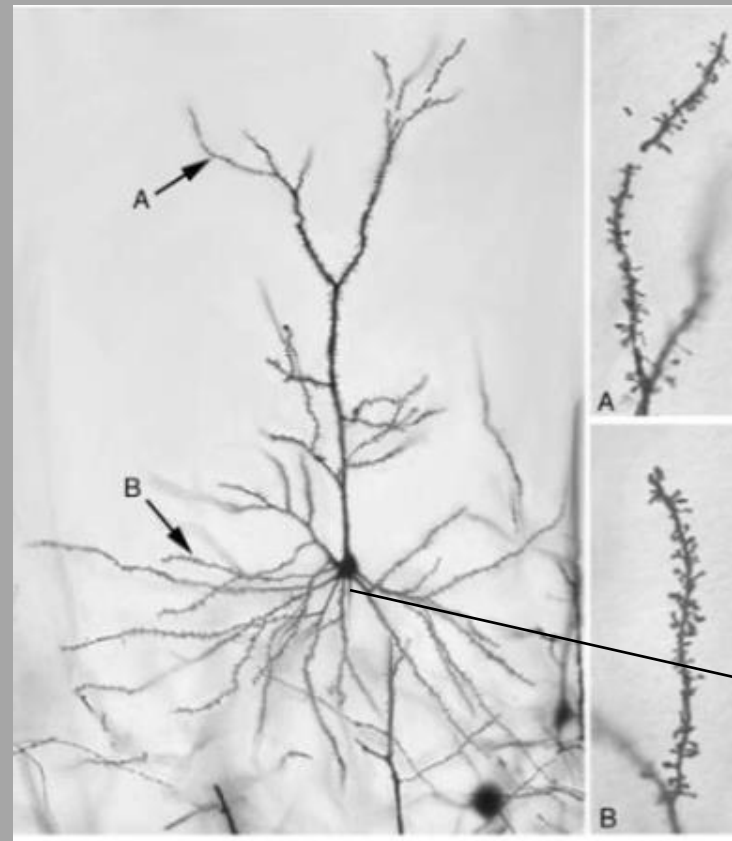
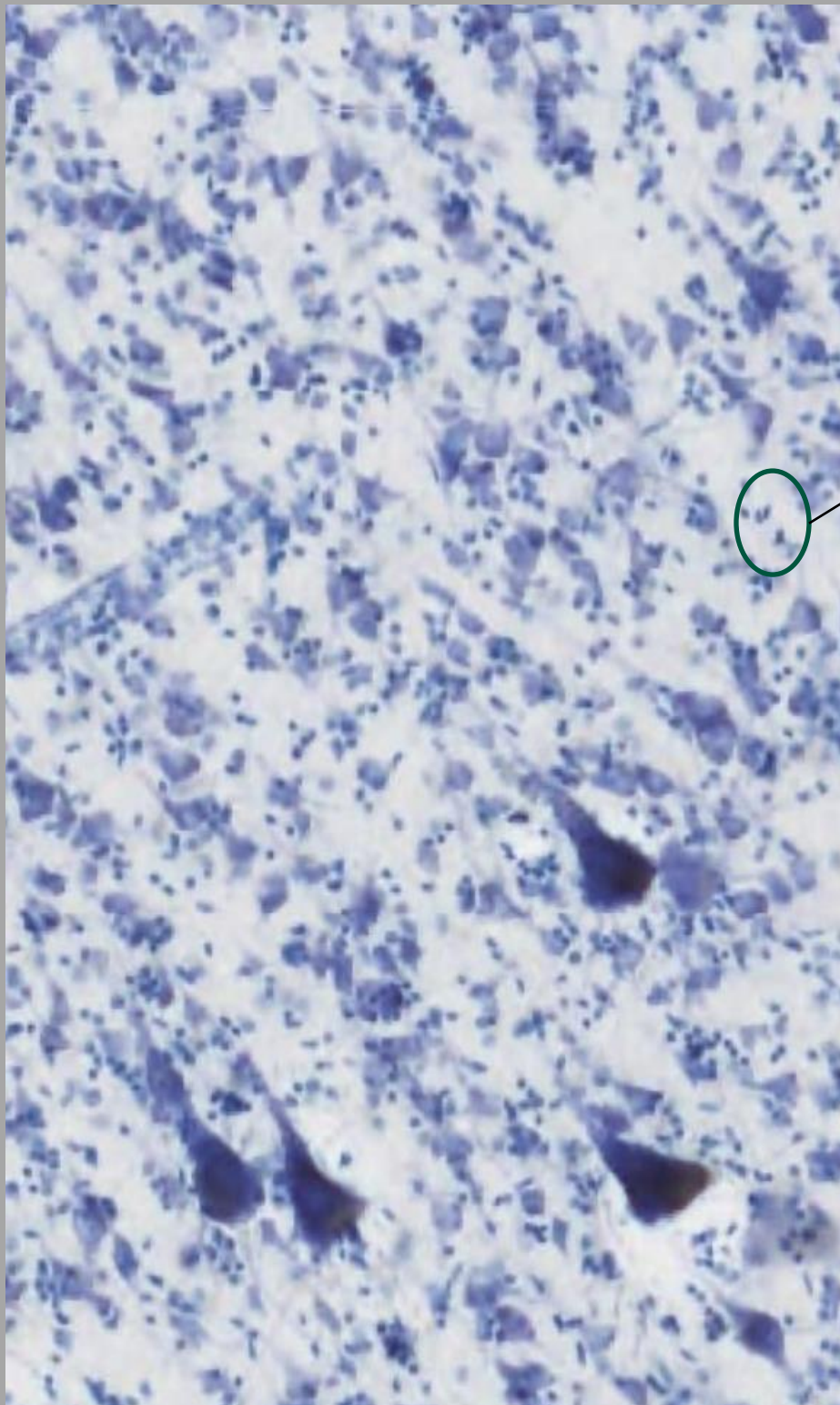
~ 85% of cortical neurons are pyramidal (Make + release Glutamate, spiny dendrites)

~ 15% are non-pyramidal neurons (Make and release GABA, non-spiny dendrites) Not seen here, need different stain

~ 75% of all cells in neocortex are glia i.e astrocytes, oligodendrocyte, microglia (Circled in picture)

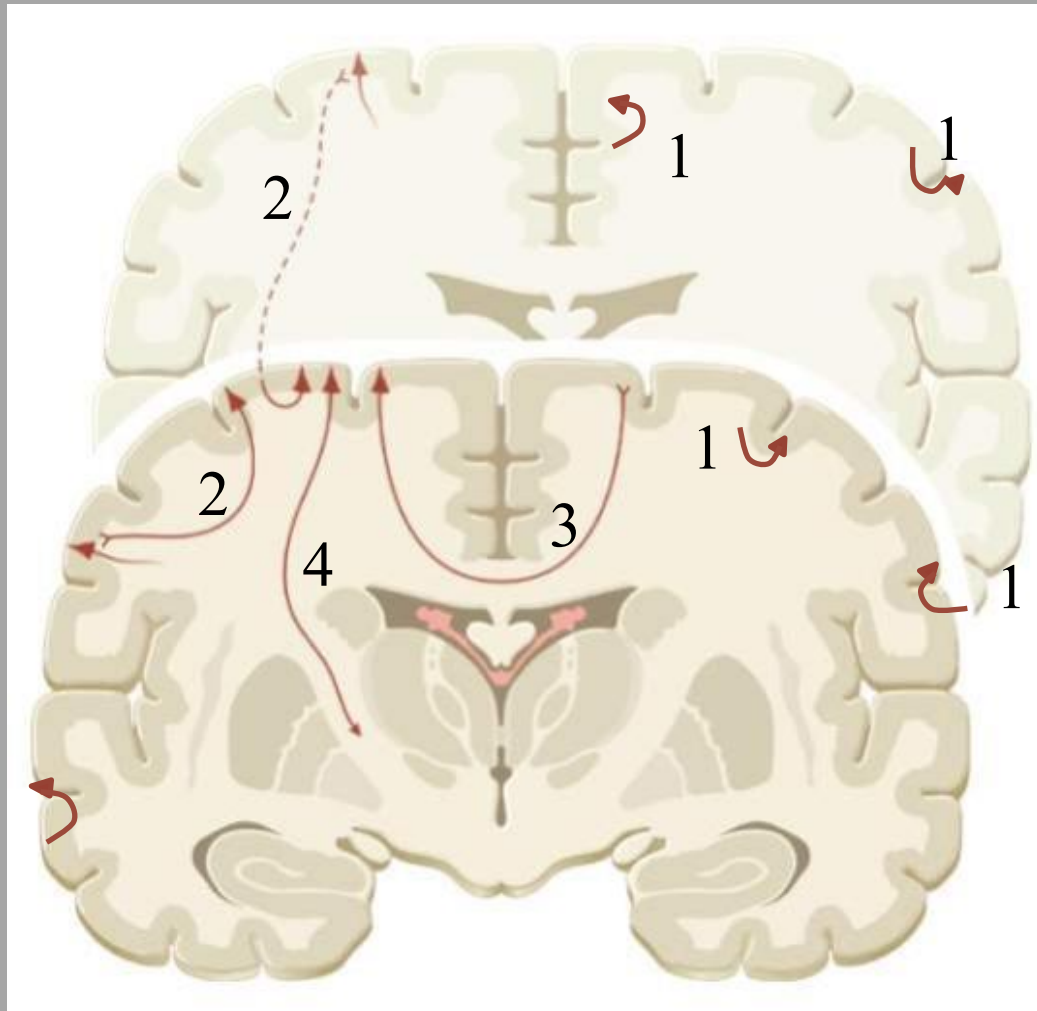
Pyramidal neuron, spiny dendrites seen

Remember parts of the neuron?



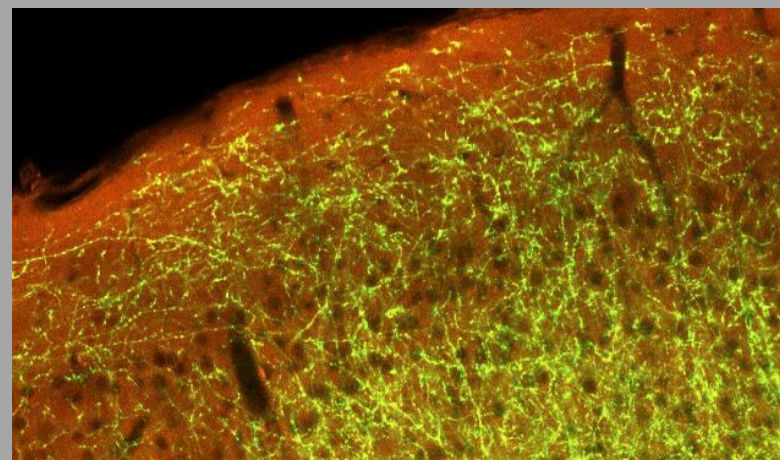
Axon going to Sub-cortical white matter

4 prototypical types of connections made by neocortical excitatory neurons:

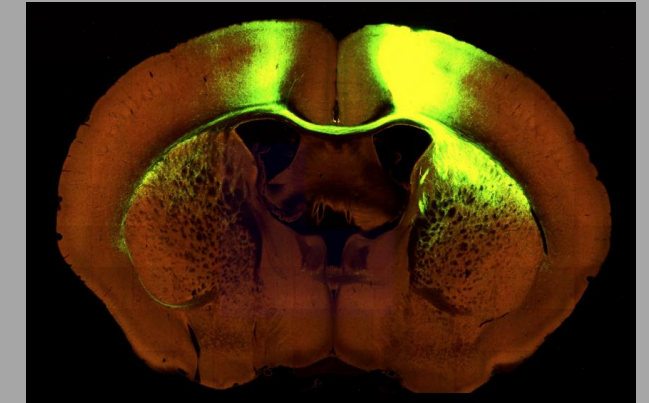


- These connections underpin the biology that drive complex human behaviors such as emotional regulation, empathy, morality, as well as sensorimotor integration etc
- Makes it very very hard to understand what neural pathways are responsible for what emotions i.e what neural pathways are aberrant in generalized anxiety disorder?

1. Local cortico-cortical
 - within one area
 - across/within layers
2. Long range cortico-cortical
 - from one area to another
3. Interhemispheric/callosal
 - generally homotopic areas in opposite hemispheres
4. Subcortical
 - thalamus, tectum, basal ganglia, limbic system, spinal cord

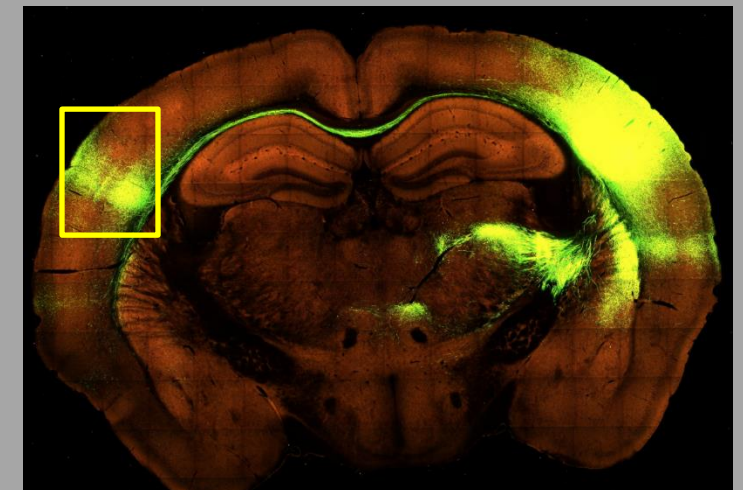


Ex 1. Motor cortex projections to contralateral motor cortex and caudate + BG

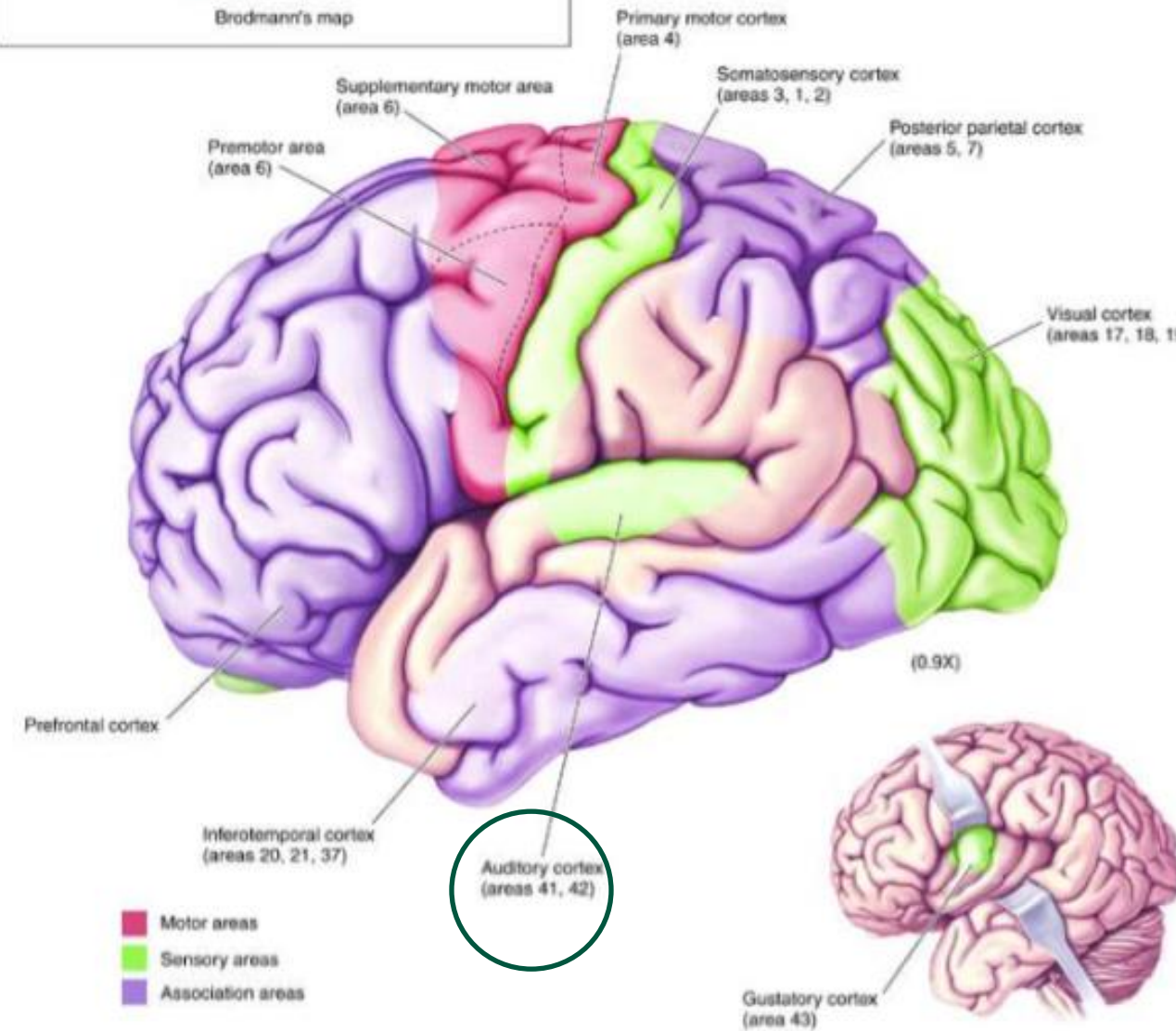
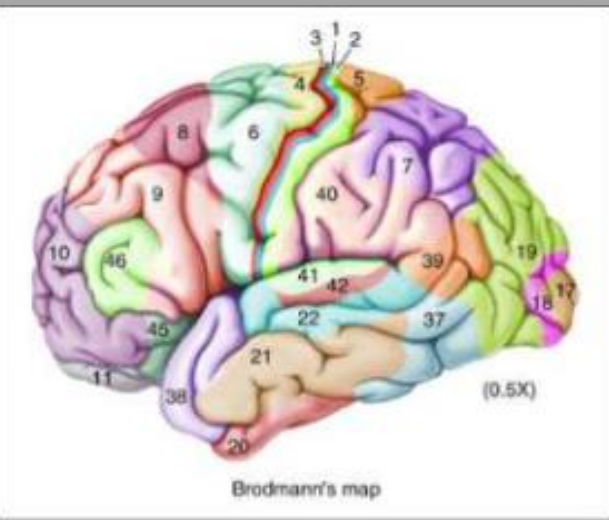


- Dye fills the neurons at the site of injection and labels the axon targets of those neurons.

Ex 2. Somatosensory cortex projections to contralateral cortex and thalamus



Understanding difference in the cytoarchitecture *** of the different lobes, gyri and sulci in various parts of the neocortex, as well as using functional maps will help one understand how the neocortex works



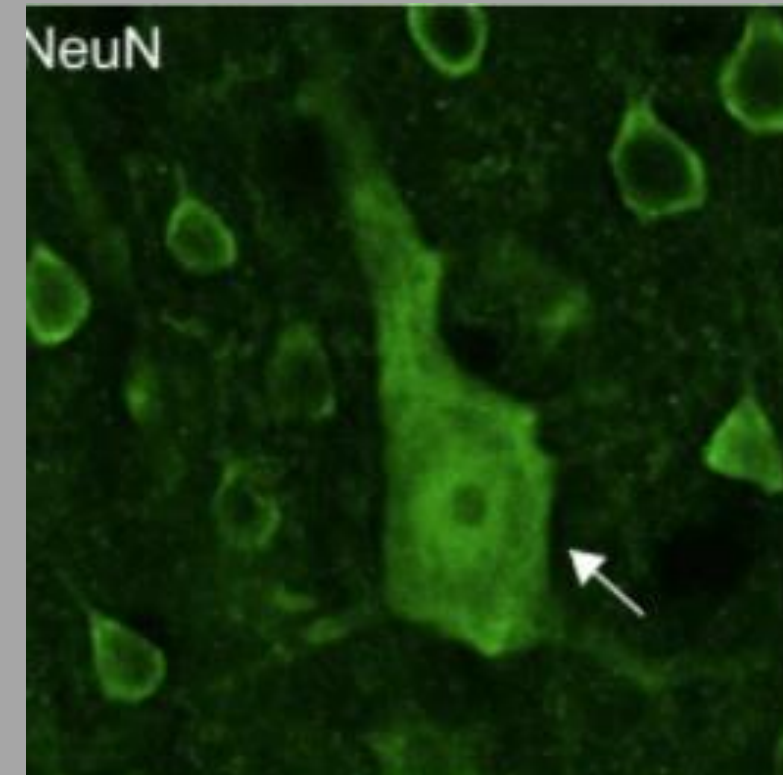
1. Major gyri are often distinct functional zones separated by major sulci.

-ex. central sulcus divides motor & somatosensory areas

2. Brodman's areas (1909) largely align with functional zones.

- ex. auditory cortex & visual cortex. About 50 areas divided based on histology of the tissue. Structure and function related, histological differences differentiate/define tissue function relatively accurately

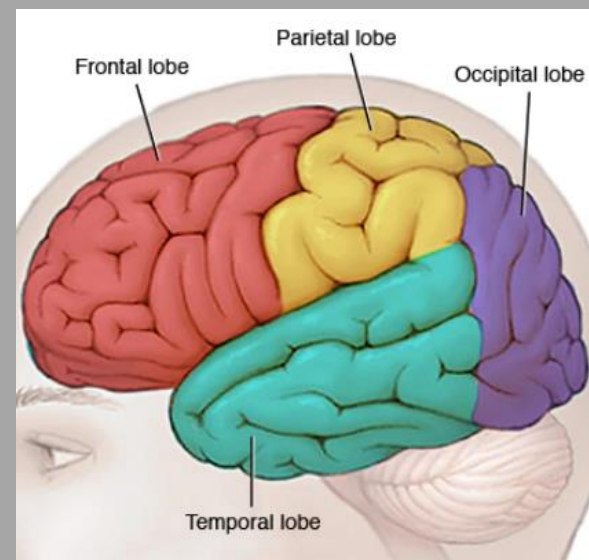
***Red arrows point to Betz neurons. Unique to layer 5b of the primary motor cortex. Send projections via corona radiata to corticospinal tract. Example of unique cytoarchitecture found in only one layer of the neocortex. Affected in ALS



Functional Neuroanatomy of the Neocortex: past & present

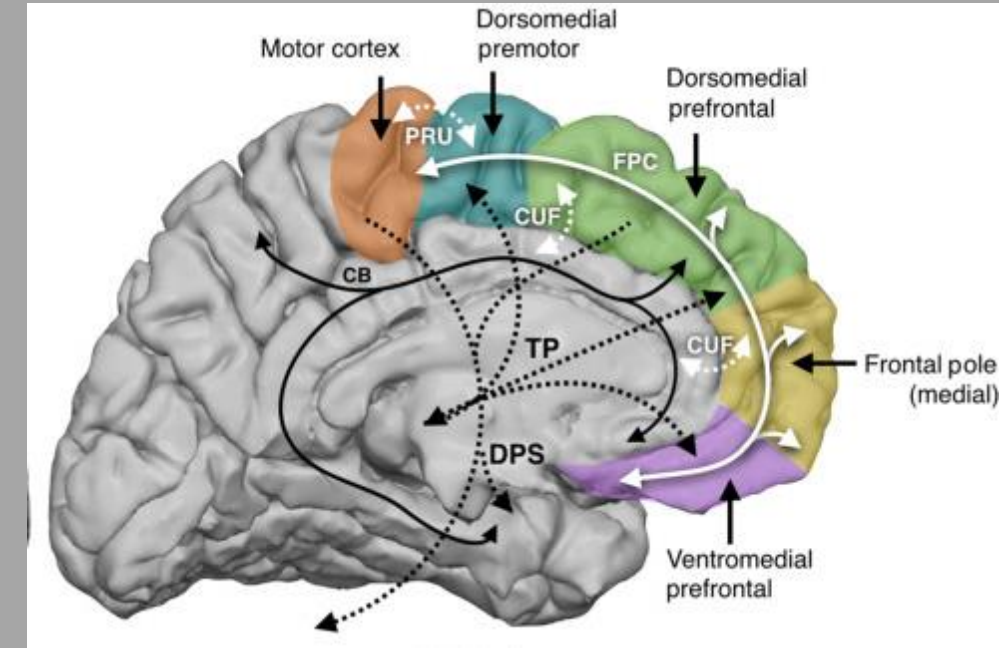
How do we learn about what the function of different parts of the human neocortex? 4 major ways

- **Clinical case reports** documented brain injury, neoplasm, vascular accident, etc. in a specific area of the brain with a correlated behavioral/clinical deficits ex) Wernicke's/Broca's area. Injury provides a clue of function
 - Requires rigorous histopathology or sophisticated imaging (MRI), not all injury or pathology is similar in extent or location among different patients (difficult to interpret)
- **Functional MRI** useful for learning about brain activity in specific regions during behavioral, sensory or motor tasks
 - Behavior limited in MRI environment and resolution is limited. Does not record single neuron, rather large groups of neurons
- **Electrical brain stimulation** coupled with behavioral observation during neurosurgery
 - Stimulation probably being performed during surgery for tumor or epilepsy treatment. Helps surgeon also to know what pathways to possibly avoid in surgery, ex) don't cut out or damage Broca's area when excising this CNS lymphoma. Limited regions sampled and not a healthy brain. Studies have caveats with this technique
- **Electrical recording of neuronal activity** during behavior/cognitive task
 - Recording probably being performed during surgery for tumor or epilepsy treatment. Limited regions sampled and not a healthy brain. Both stimulation and recording can be done before a surgery to remove some lesion

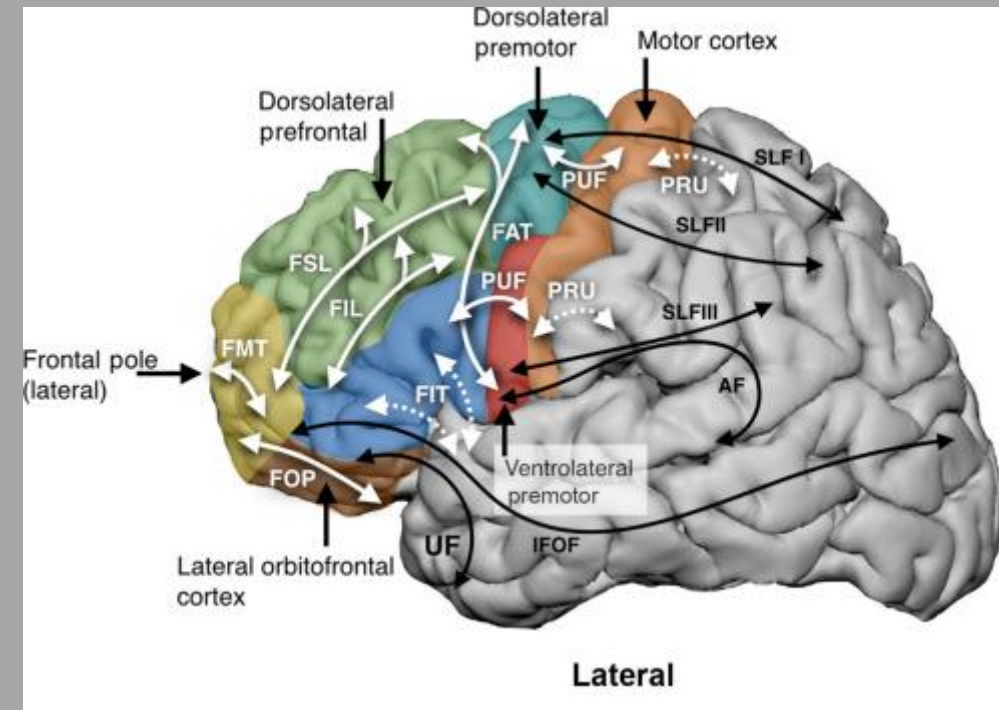


Frontal lobe/PFC: executive & voluntary motor function including speech, emotion, cognition.

- Executive functions (EF) are a collection of distinct but related cognitive functions that typically refer to aspects of cognition and behavior required to successfully **execute complex tasks** (Denckla, M. B. 1996).
- Frontal lobe/ PFC functions include but not limited too: **setting and maintaining goals, planning, delayed gratification, organizing, initiating, inhibiting, self-monitoring and evaluating, thinking and acting strategically, problem-solving, shifting flexibly among cognitive and affective sets, deliberately controlling cognitive functions, and managing moment-to-moment events.**
- Behavior at work vs with family
- A **cognitive set** governs a person's thinking and information processing, while an **affective set** guides their emotional responses
- Also remember the pre-motor cortex and supplementary motor area are in the frontal lobe, which means frontal lobe has a role in planning, organizing and executing complex motor tasks



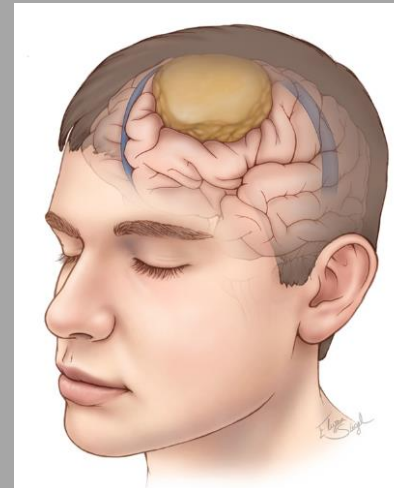
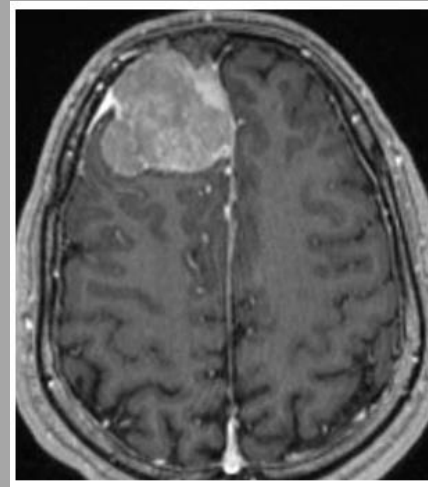
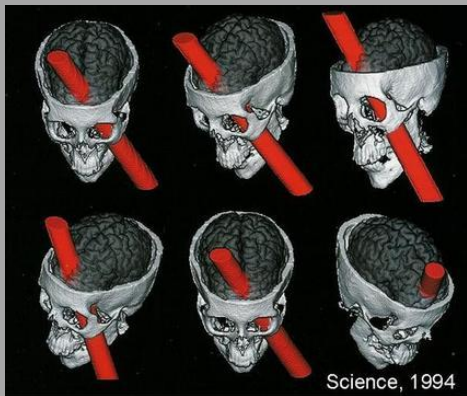
All cortical areas are connected to numerous other cortical areas as well as subcortical areas



Frontal lobe, mood and psychiatric disease

Psychiatric disease is associated with changes in frontal lobe function & neuroanatomy

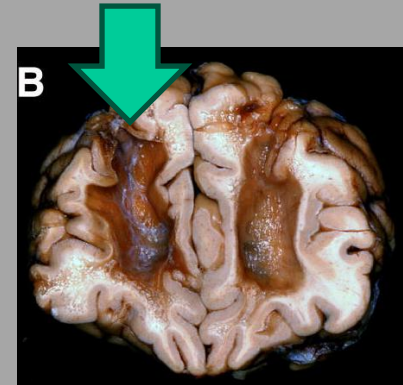
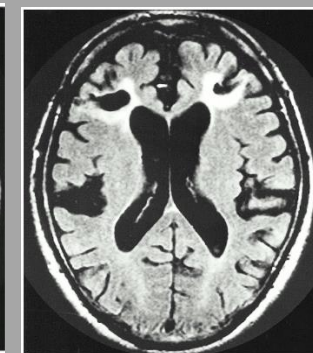
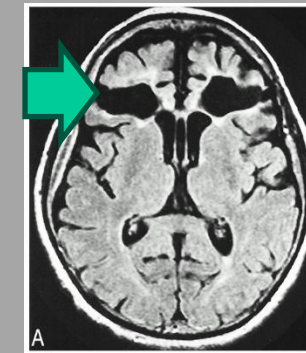
- Phineas Gage 1848 had an iron rod pierce through his orbitofrontal cortex (PFC). Before the incident, Gage was a loving husband and father. After the incident became short-tempered, brash, and prone to devise and then immediately abandon any plans he made.
- Great case study of frontal lobe injury leading to changes in personality, affect, social behavior, etc.



Frontal lobe meningiomas or other tumors can be associated with depression, schizophrenia or other psychiatric disorders.

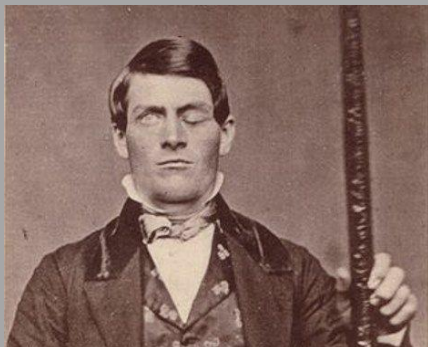
<https://www.aaroncohen-gadol.com/en/patients/meningioma/overview>

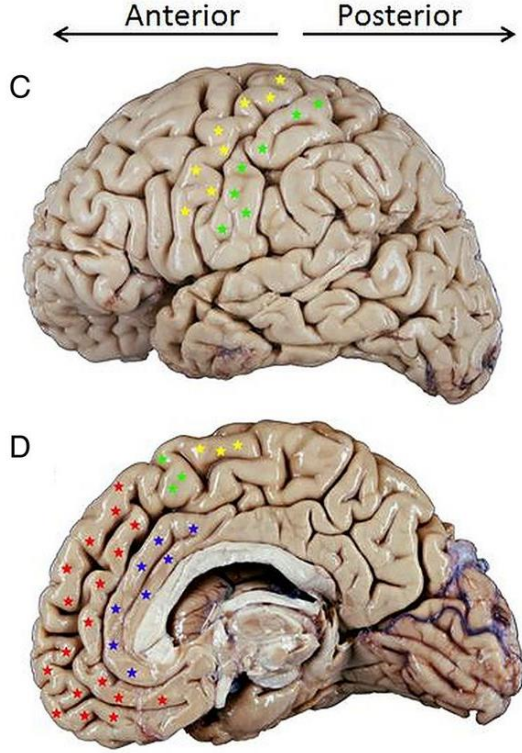
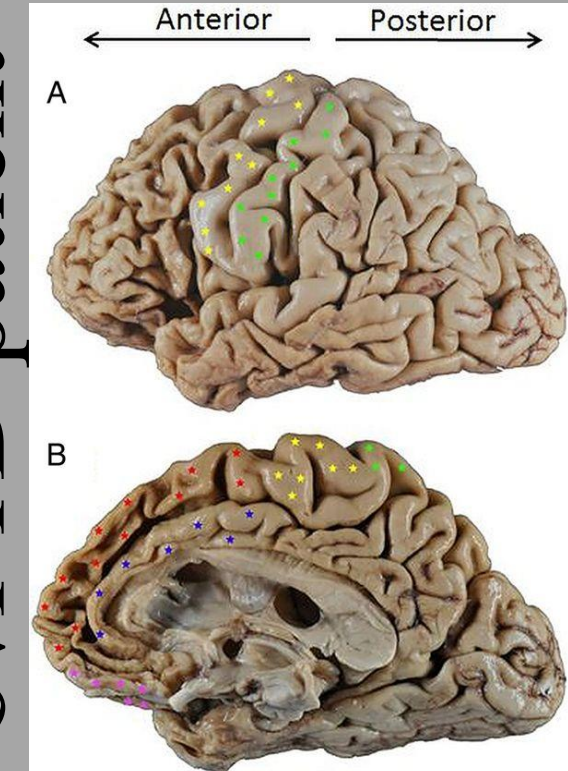
- Frontal lobotomy developed in the mid 1930's to treat psychiatric disease (eg schizophrenia) by severing connections to frontal lobe. MRI of 2 patients shown below.



- Procedure is very effective in blunting mood and affect and relieving clinical symptoms esp. in suicidal patients. No longer in use due to advances in pharmacotherapy and ethical issues, mortality, etc.

Uchino et al. 2001 Cranial MR Imaging of Sequelae of Prefrontal Lobotomy
Szczepanski and Knight 2015





SC Lanata & BL Miller 2016

bvFTD; behavioral variant FrontoTemporal Dementia/Degeneration

Deficits in Executive Function

Poor decision-making, judgment, problem-solving, and organizational skills. Examples include: Difficulty planning the day's activities, Questionable financial decisions, On-the-job mistakes.

Disinhibition: A loss or lack of restraint based on social norms, leading to inappropriate behavior and impulsivity. Behaviors may include:

- Making uncharacteristic rude or offensive comments → Boards questions will usually include this in stem
- Ignoring other people's personal space, reckless spending
- Touching strangers or inappropriate sexual behavior
- Aggressive outbursts or Apathy
- Onset is usually between 45-65 years old. Younger than Alzheimer's Disease

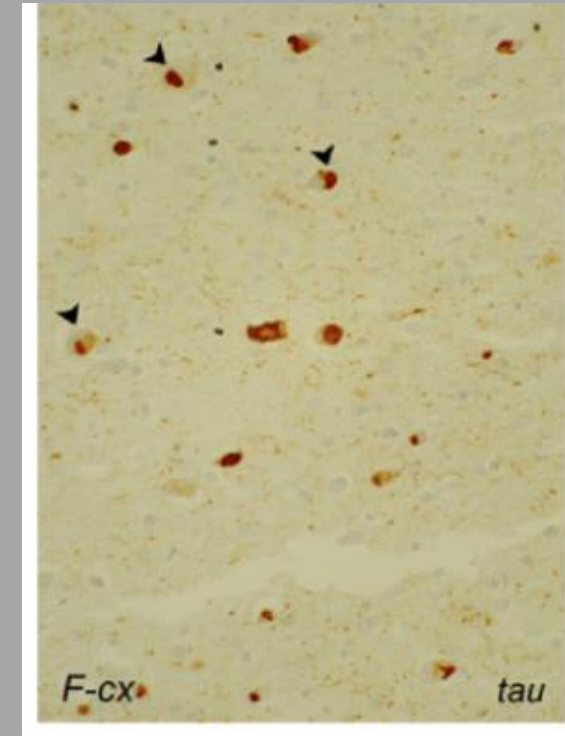
Indifference or lack of interest in previously meaningful activities. Behaviors may include: Loss of interest in work, hobbies, and personal relationships, Neglect of personal hygiene.

Loss of initiative, Emotional Blunting, Loss of warmth, empathy, or concern for others. Behaviors may include: Indifference to important events (e.g., death of a family member or friend); Failure to recognize that loved ones are upset or unhappy, Compulsive or Ritualistic Behaviors, Single behaviors or routines that are performed over and over.

Adapted from <https://www.theaftd.org/>

Caused by mutations in MAPT - tau

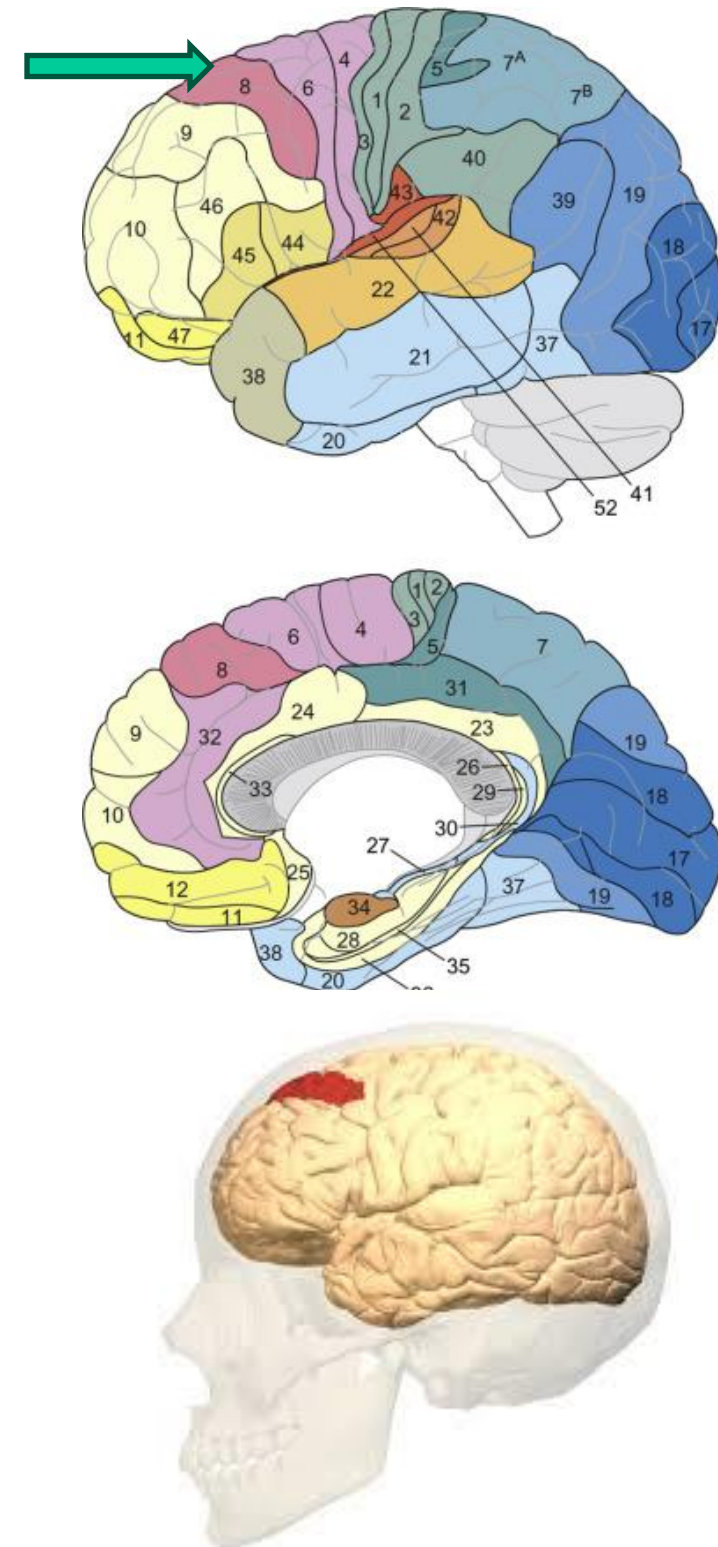
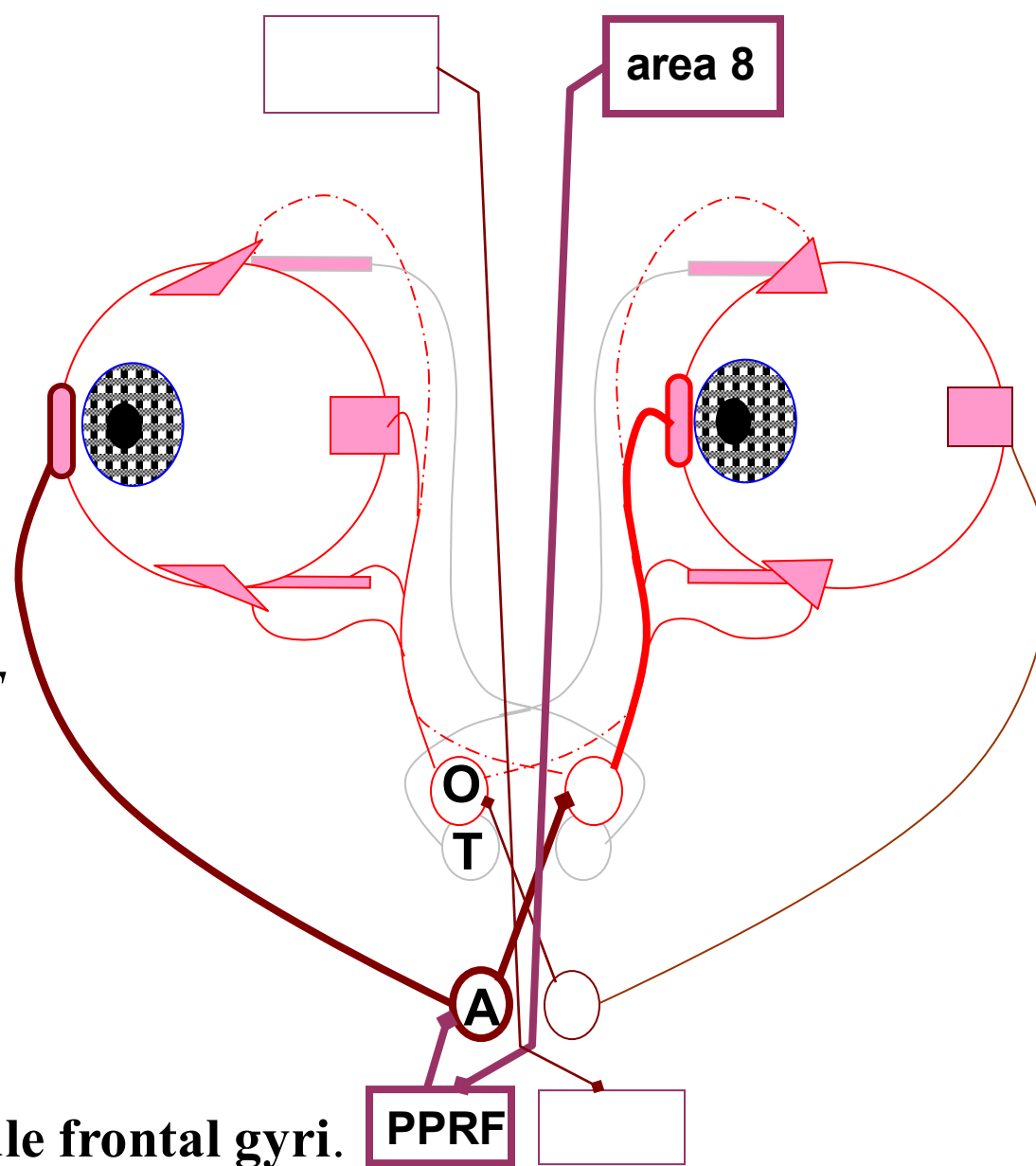
Tau aggregates



Frontal lobe area 8

- Vital for rapid, saccadic voluntary eye movements

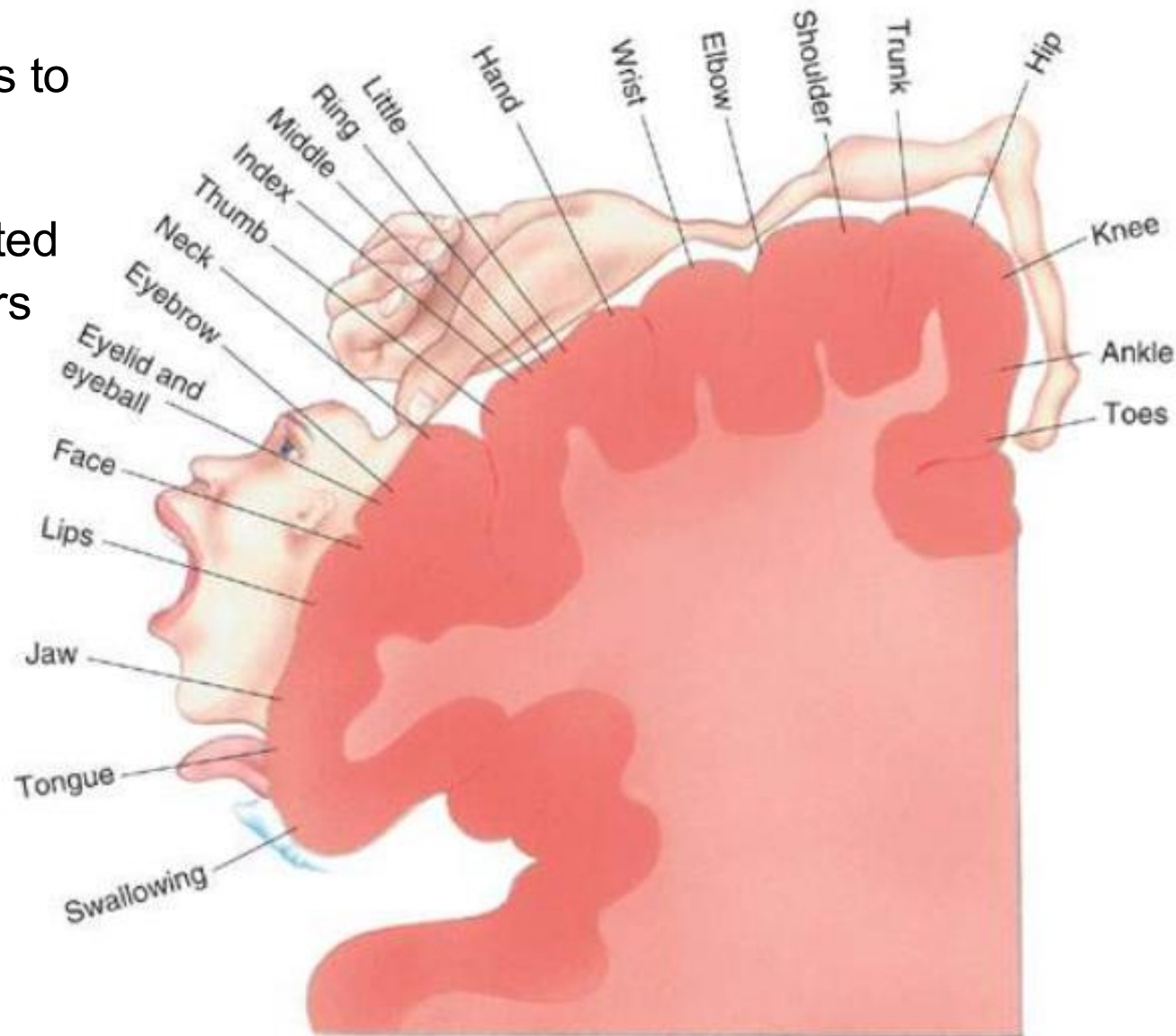
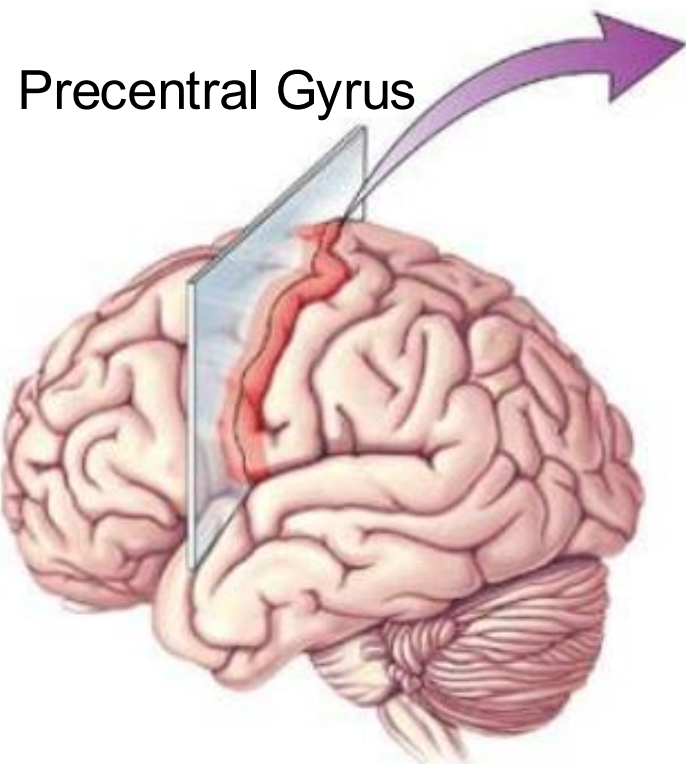
- *Cortical area 8, Sends projections to brainstem (Paramedian Pontine Reticular Formation) → PPRF then projects to abducens and oculomotor nuclei to carry out eye movement*



- Found in parts of **superior and middle frontal gyri**.
- Receives inputs from visual cortical areas.
- Electrical stimulation causes eye movements.
- Electrical recordings show action potentials in area 8 neurons before eye movements.
- Area involved in imitation/planning of the eye movement
- Lesion/injury to Area 8 cause deficits in voluntary eye movement

Primary Motor Cortex

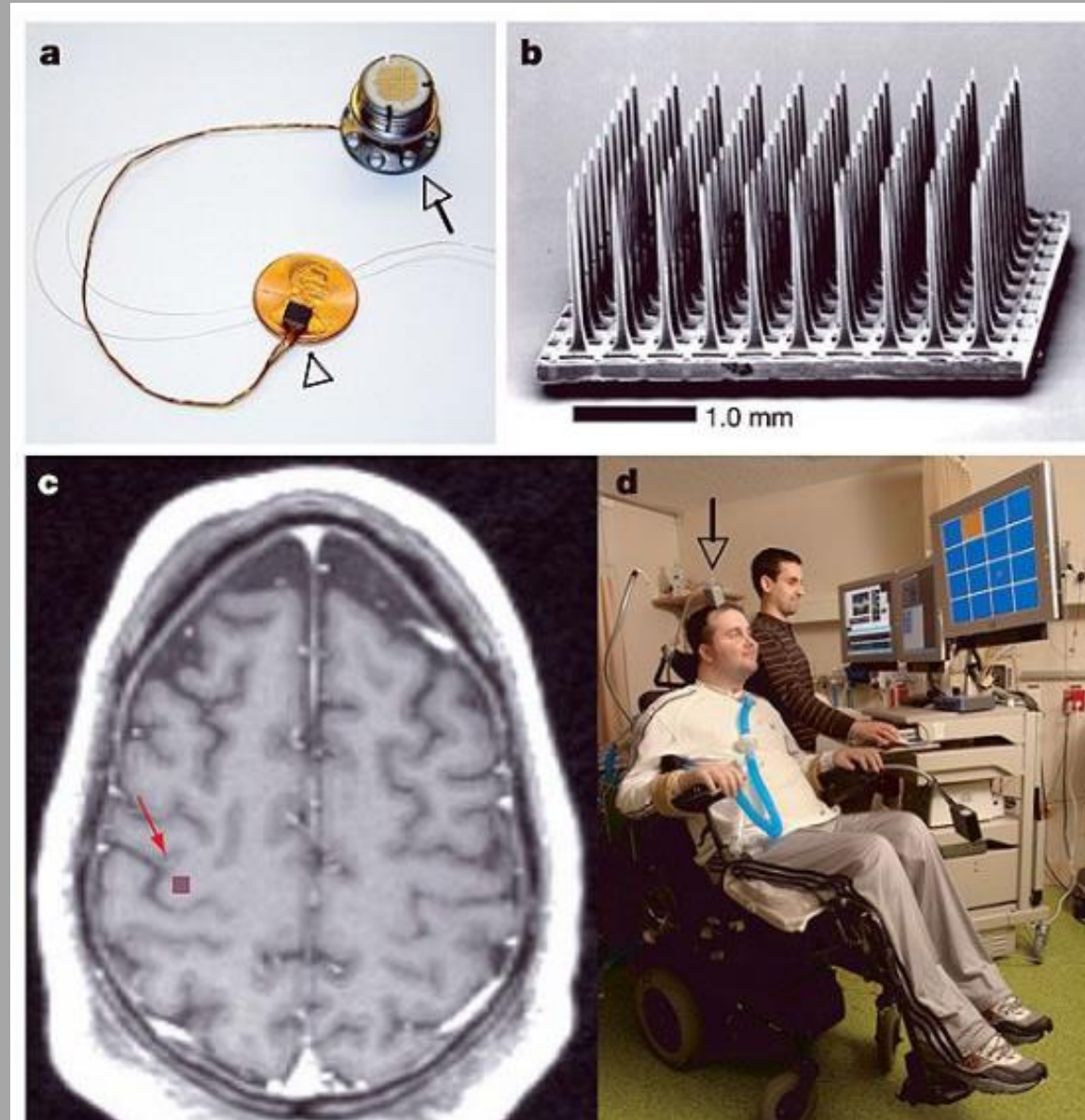
- Produces voluntary motor movements via projections to spinal cord
- Many more neurons devoted for movement of the fingers and face musculature.



- Afferents from motor, vestibular, sensory systems. i.e catching a football with your hands or catching yourself from tripping down the stairs
- Efferents to thalamus, brainstem, spinal cord, basal ganglia for voluntary motor movements.
- Electrical stimulation causes muscle twitches.

Control of prostheses with neuronal activity from primary motor cortex

- Intact motor cortex with a spinal cord injury → encode neuronal activity from motor areas and put that information into a computer to control a protistic limb to bypass the injured spinal cord.
- This can help patients who have spinal cord injuries

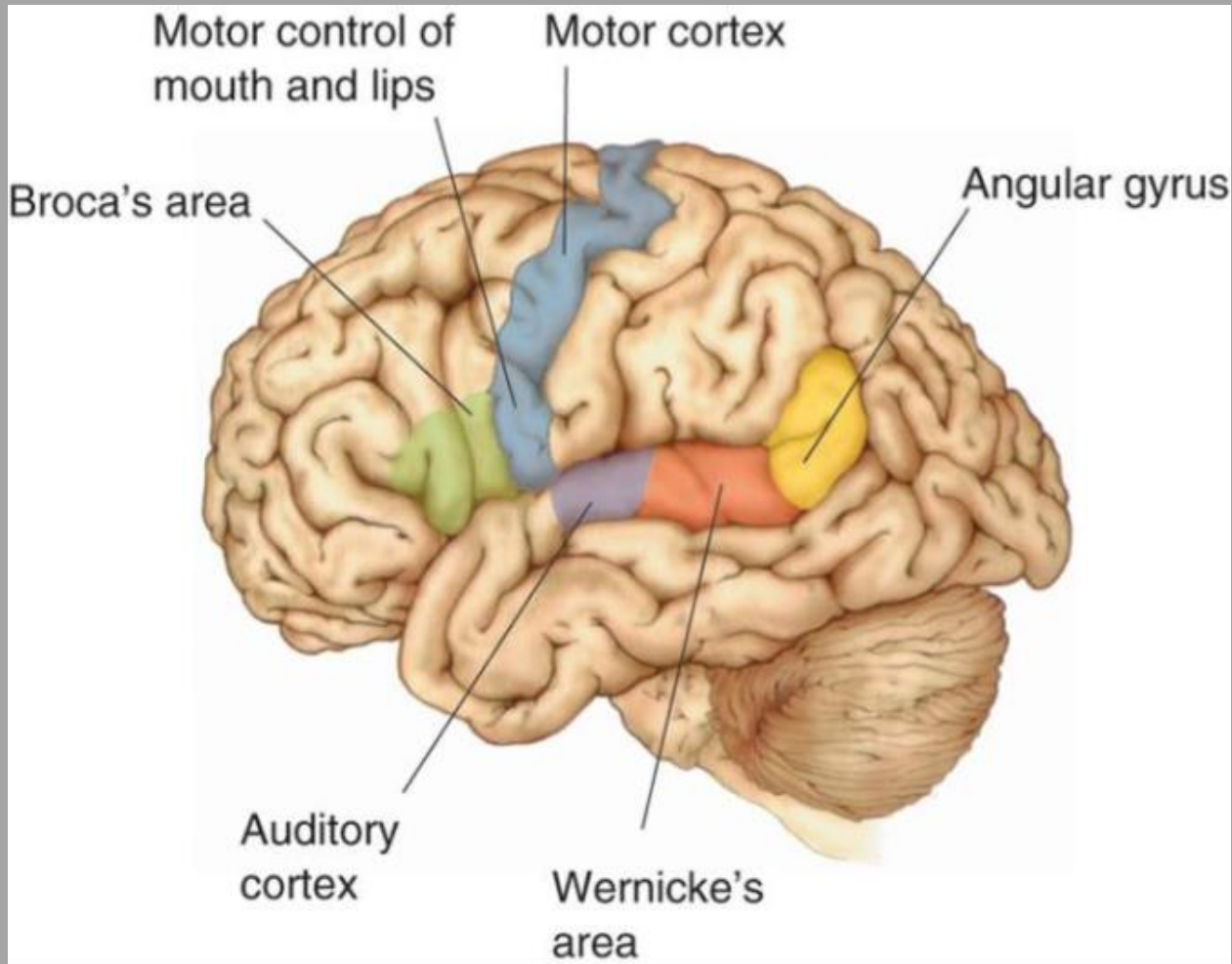


Video

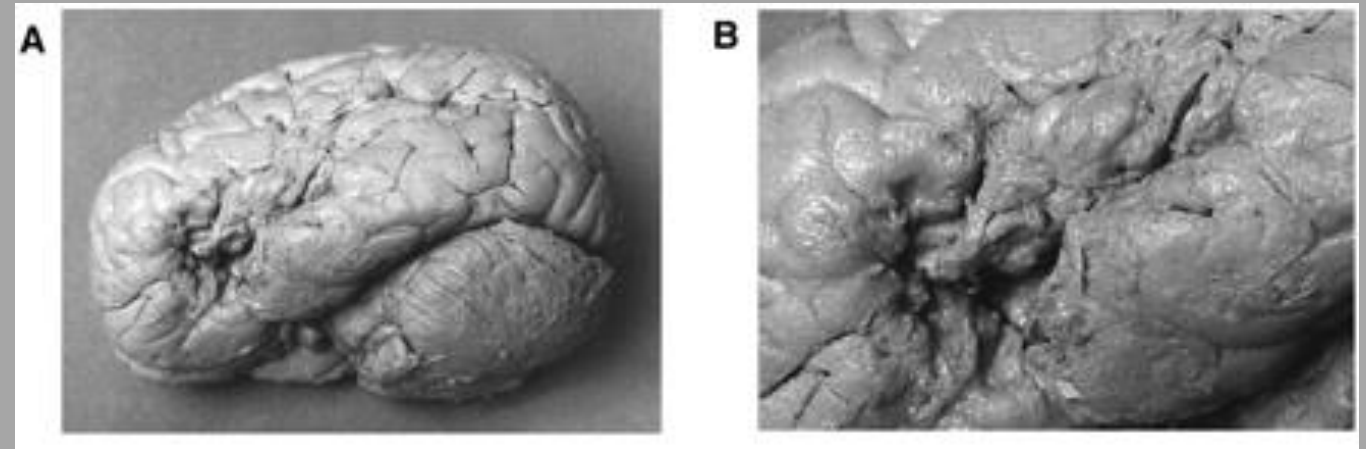


<https://www.nbcnews.com/health/health-news/paralyzed-woman-gets-robotic-arm-she-controls-her-mind-flna2910144>

Cortical anatomy of speech production



- Named by Paul Pierre Broca. Broca's area in frontal lobe. Broca's aphasia (Pierre 1861): difficulty with production of spoken language (also called 'expressive aphasia').



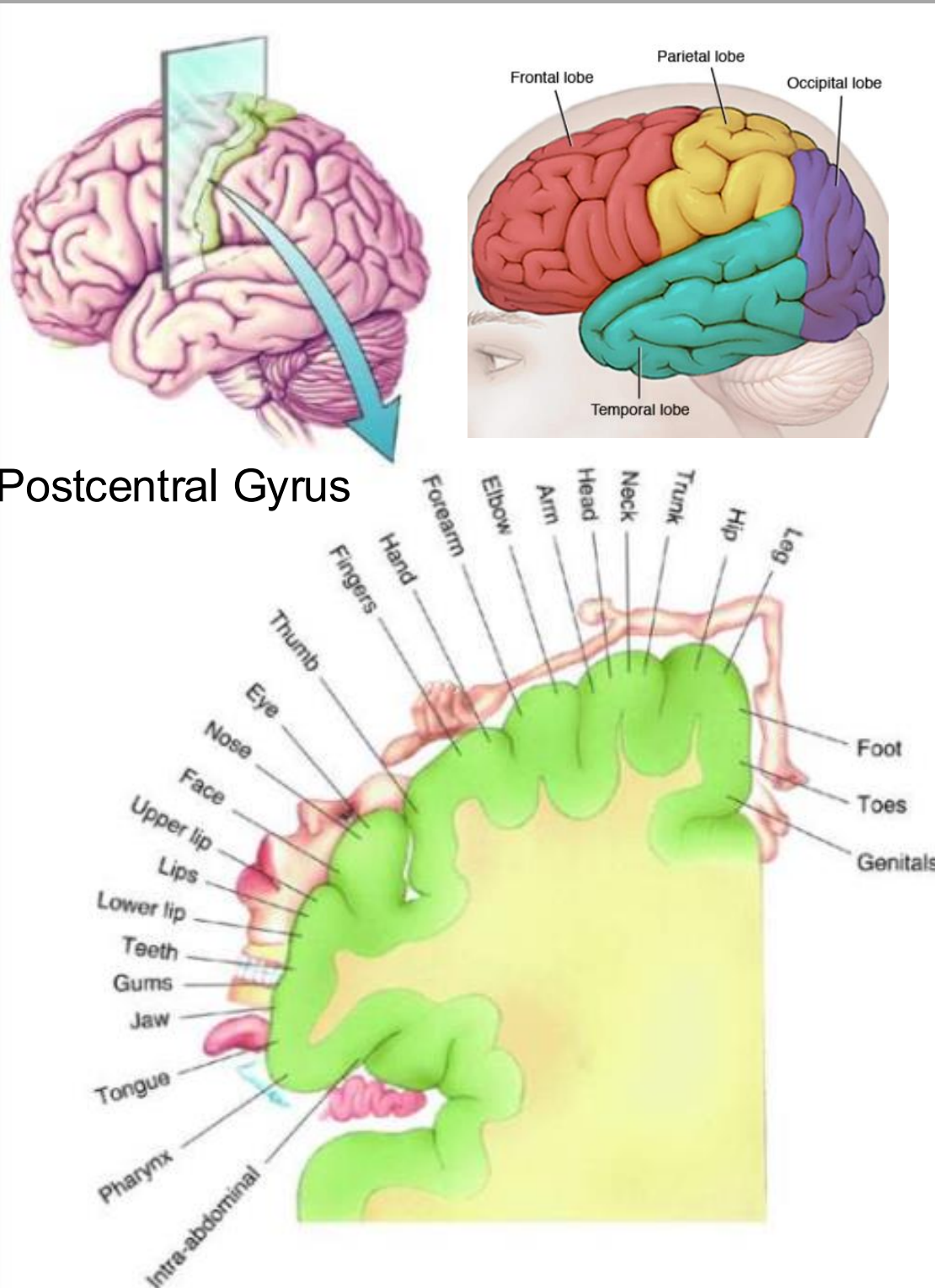
Receives inputs from auditory and motor areas.

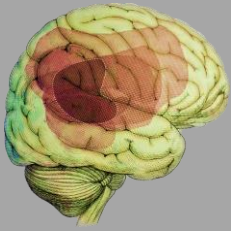
Parietal lobe:

- Contains Primary Somatosensory cortex (S1)
- Higher order sensory processing (visual-spatial attention)

Postcentral Gyrus

1. Somatotopy – map of dermatomes
2. Many more neurons devoted for sensation of the skin on the fingers and face.





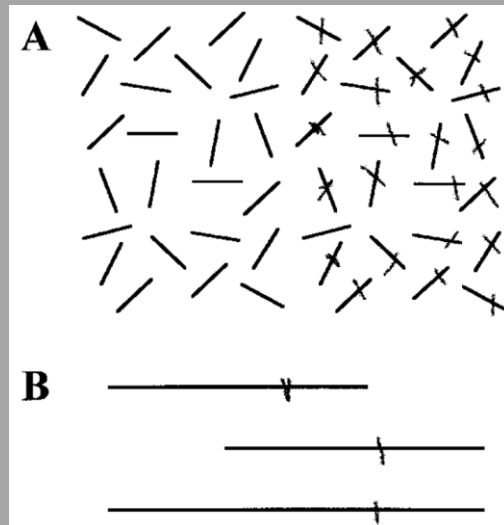
Right Posterior parietal lobe damage: Spatial Hemineglect

- Patient asked to copy of model picture shown left.
- Pts try to copy from memory, but they draw the clock/house/etc that is missing the left hand side of the images.
- Hemi-neglect: Lack of **attention** toward one side of the body & one side of the world. Will put clothes or makeup on one side of the body. **Not a vision problem.** This is an issue of higher-level processing of the afferent incoming sensory stimuli.
- Usually seen with large lesions/ injury to **right (usually non dominant) posterior parietal lobe**. Artery supplying this brain region is the Right MCA. Right posterior parietal lobe involved in spatial processing for both left and right sides.
- Right posterior parietal lobe seems to play role in visual appraisal/ attention to world around us. Primary visual cortex has projections to parietal lobes. (Another lecture)
- May present after a R MCA stroke

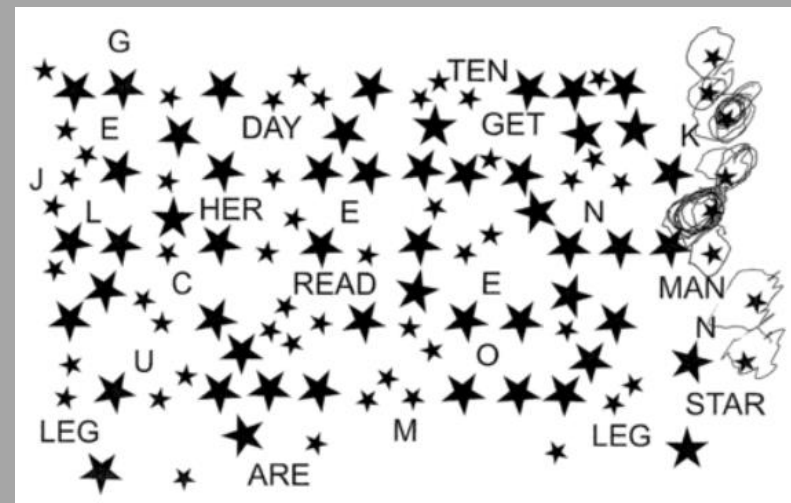
****All the tests shown are administered by putting the paper right in the middle of their face so that left and right visual fields can be assessed. When this is done, the patient only fills out one side of the paper, likewise in the crossing the lines task or the circle the stars task.**



- Patient asked to draw line through middle of lines.
Left hemineglect

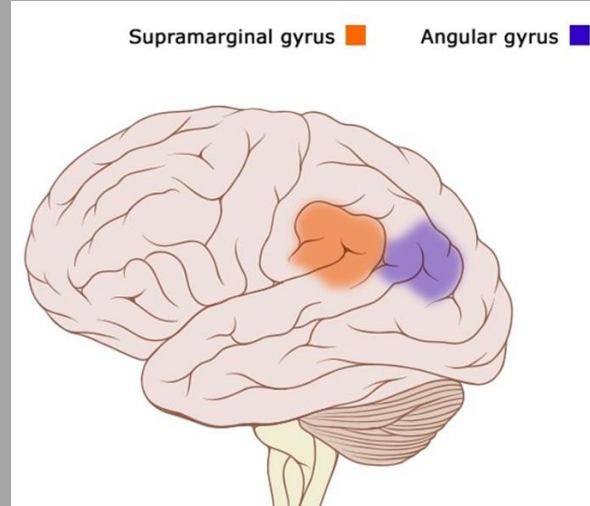


- Patient asked to circle all small stars.

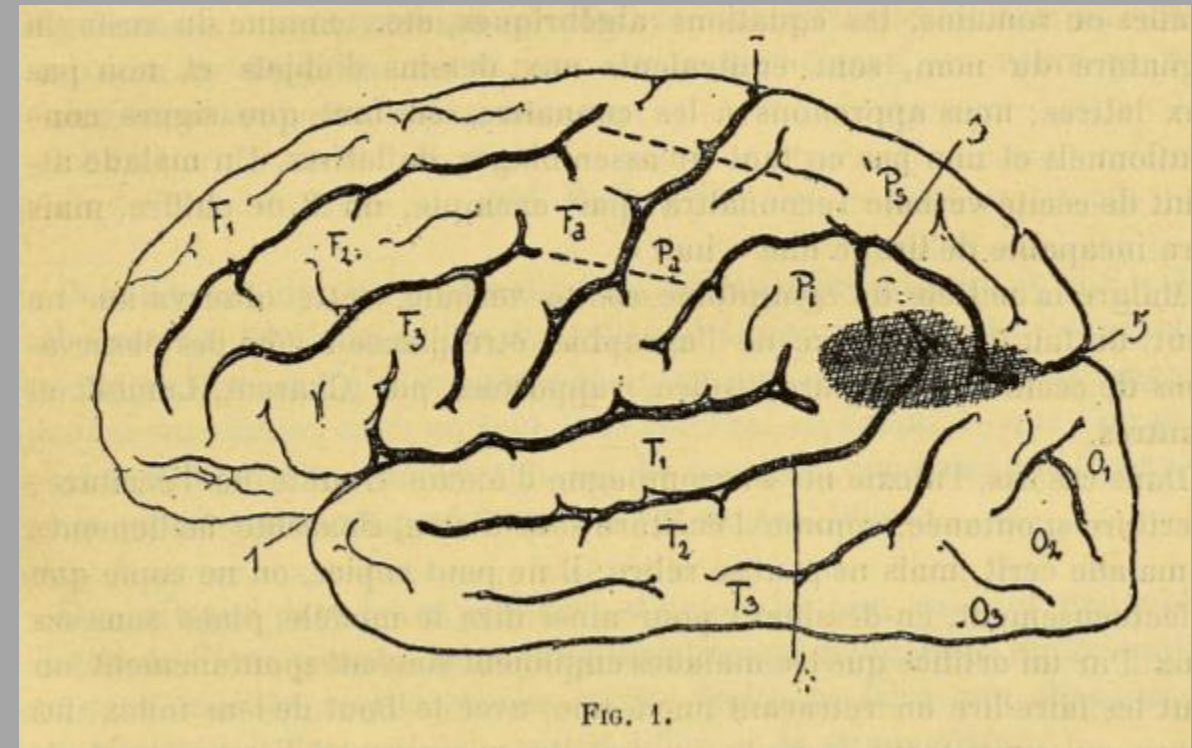
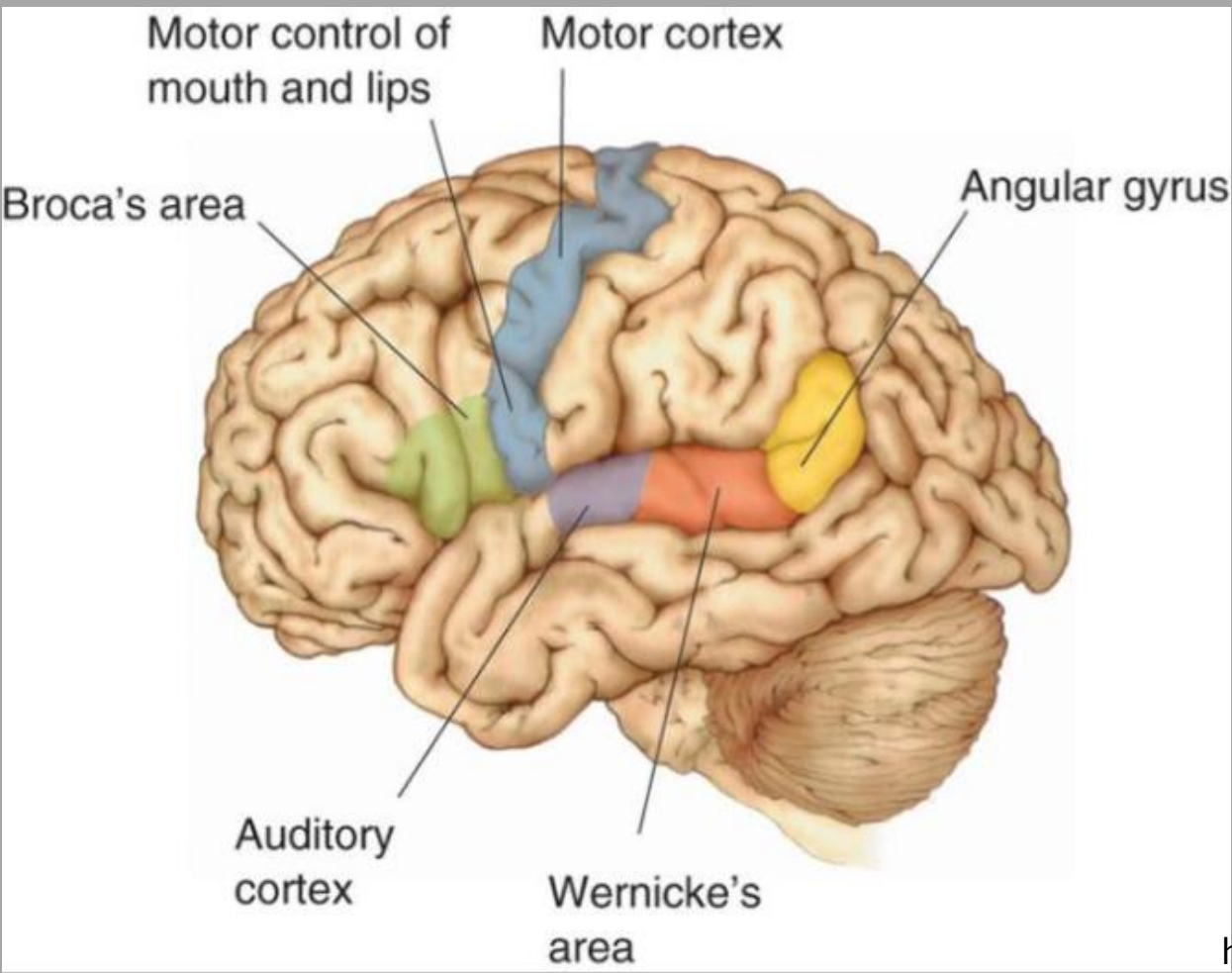


Parietal Lobe: Angular Gyrus

- Lesions to Angular gyrus (part of inferior parietal lobule) lead to agraphia w/ alexia: deficits in writing (agraphia) & reading words (alexia). Vital area for assigning meaning to words
- Angular gyrus is a heteromodal association area. Meaning it integrates sensory information from visual, auditory as well as sensory areas. Thus, it helps link areas of the brain associated in memory, spatial cognition and language together.
- Angular gyrus shows increased activity in fMRI during reading and writing
- May also see some math calculation issues → see next slide



- Anatomist Dejerine 1891: deficits in writing & reading words (can also affect numbers)

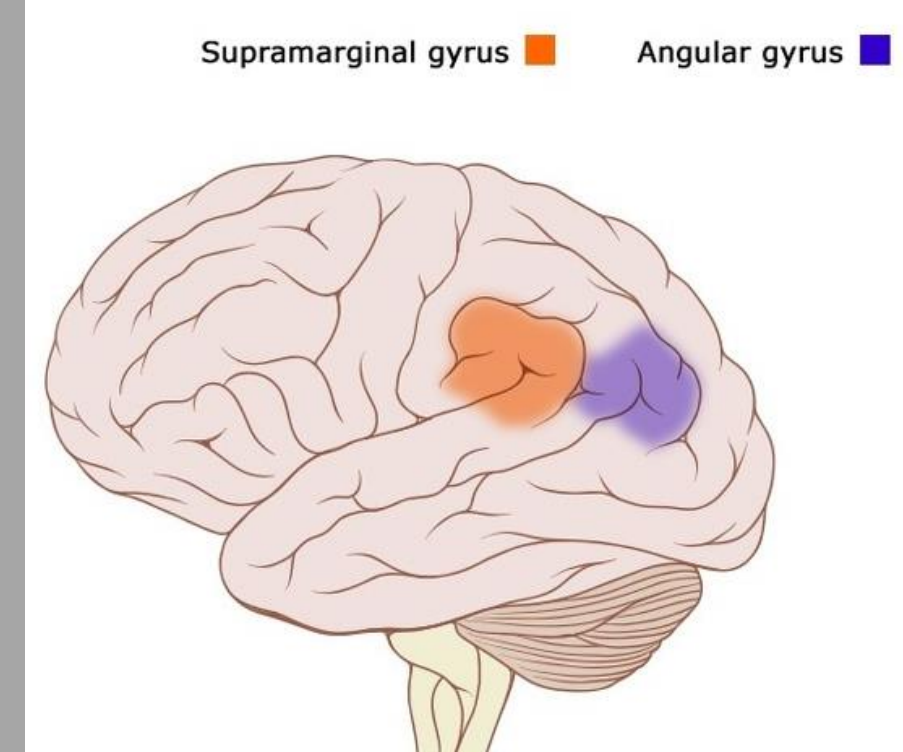


Gerstmann's syndrome:

is a dominant hemisphere stroke syndrome affecting the left (inferior) parietal lobe in the region of the **left angular gyrus**, consisting of four clinical signs:

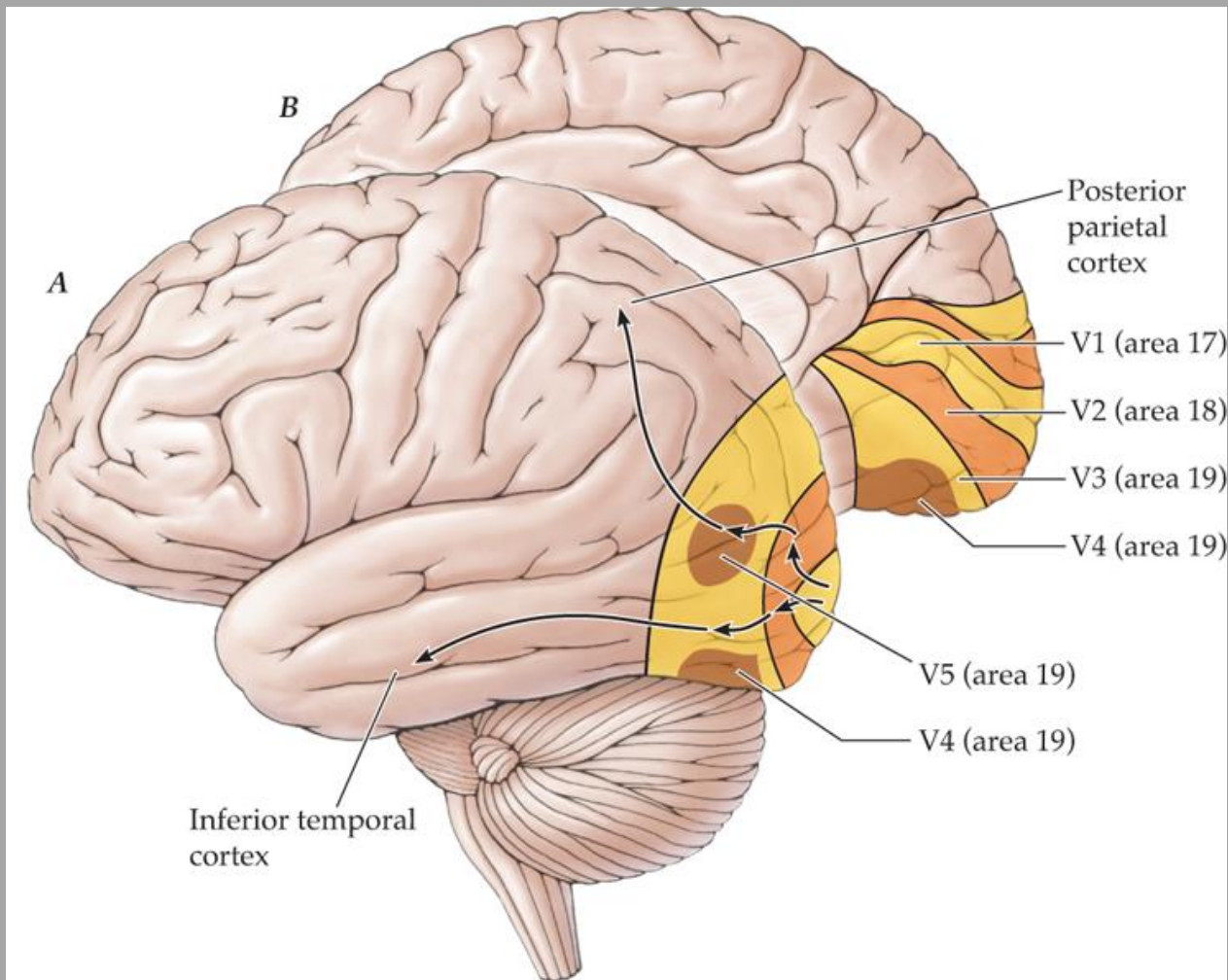
- Dysgraphia/Agraphia → writing deficits
- Dyscalculia/Acalculia: impaired arithmetic calculation abilities
- Right-left disorientation: difficulty identifying right vs left side of body
- Finger agnosia: inability to name or identify individual fingers

Spatial/
attention
defects



Occipital lobe: visual function

- Arrows show how visual information created in visual cortex can project to areas in parietal and temporal lobes involved in higher order functions related to visual inputs.



- Occipital lobes projections to higher order processing areas
- Creation of image in the occipital lobe, appraisal/processing occurring elsewhere
- Much more info in other lectures



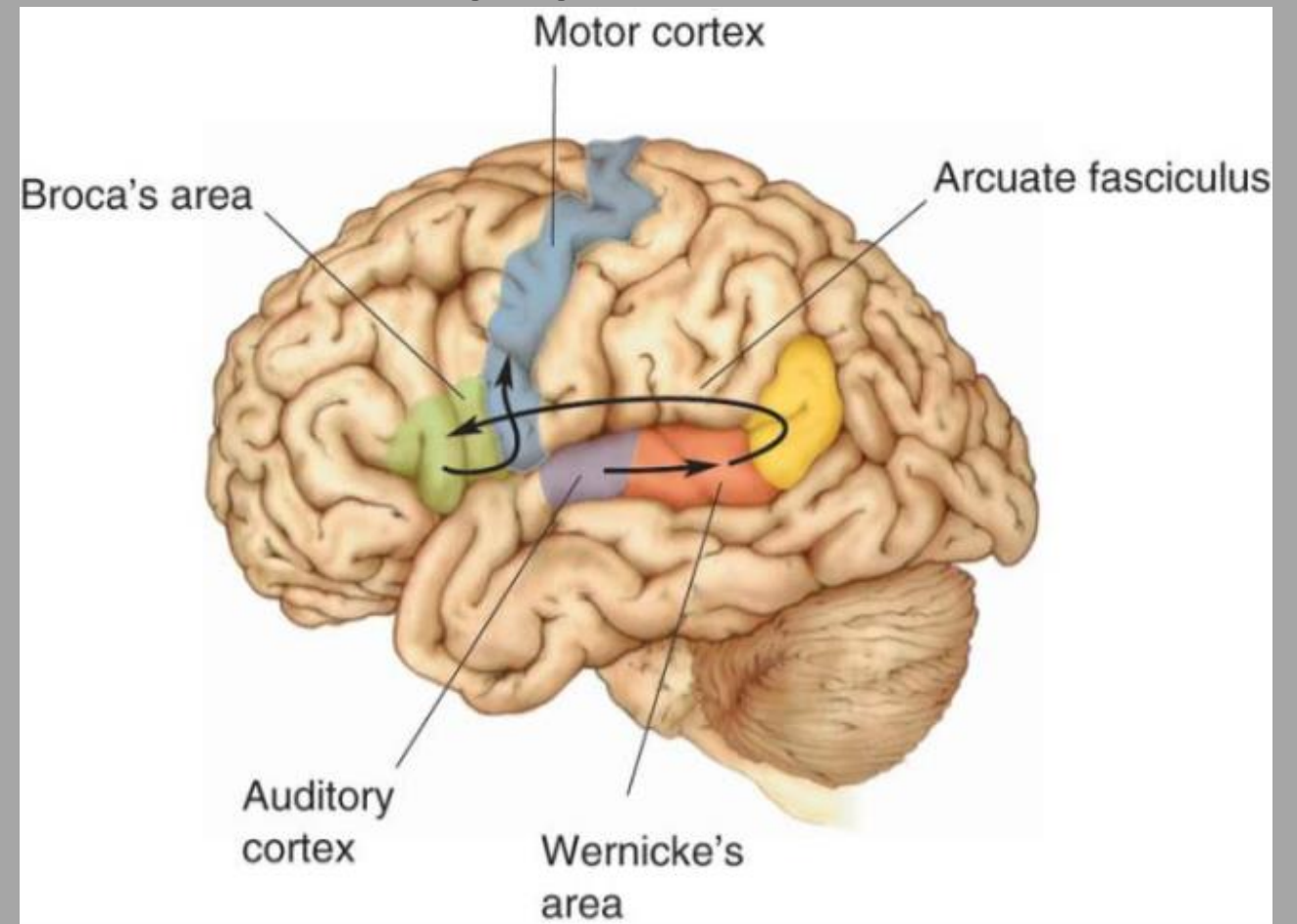
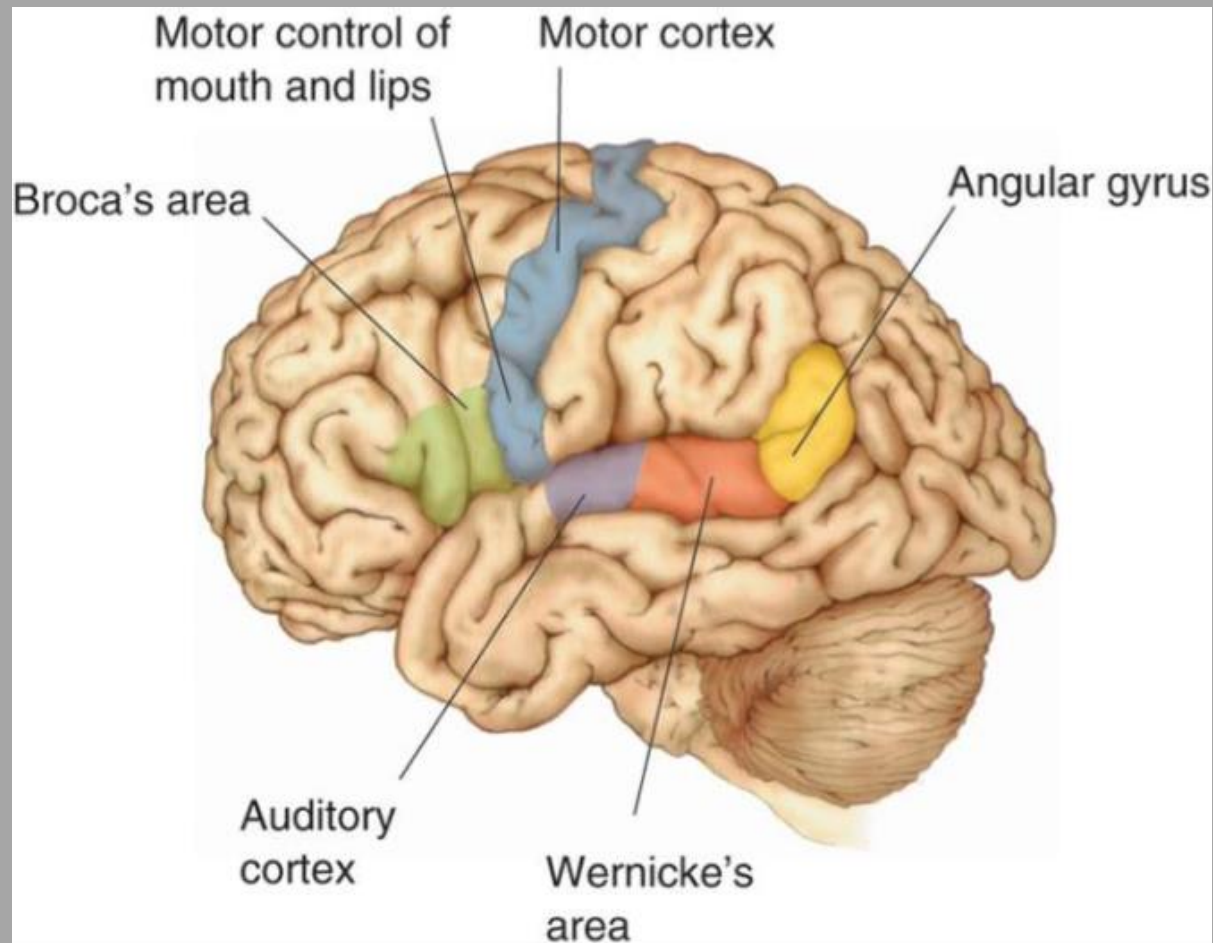
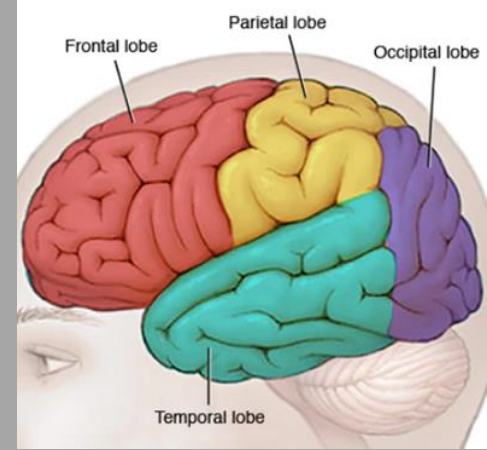
Shape
Size
Location
Orientation
Spatial frequency
Color
Movement...

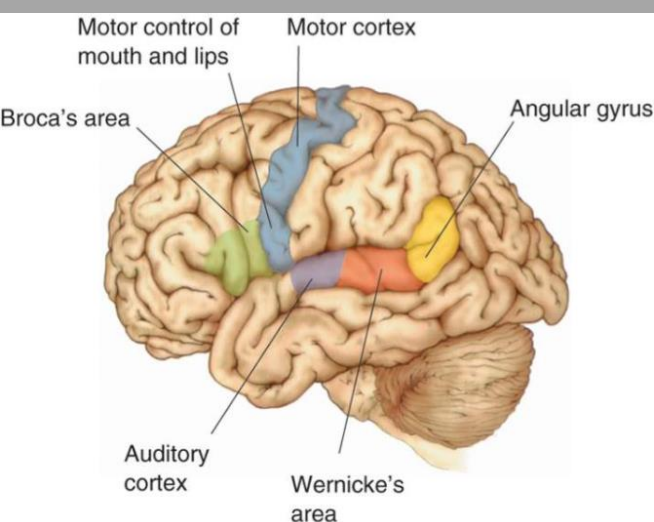
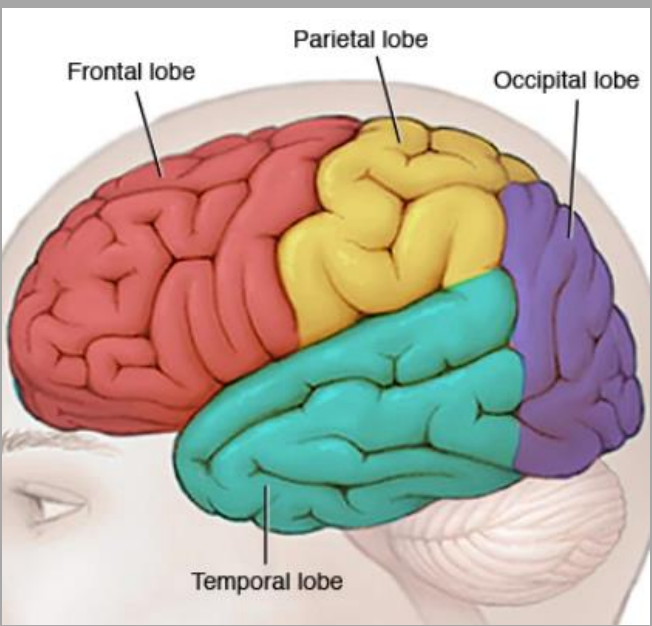
topic of future lecture...

Temporal lobe:

Audition & memory, higher order visual processing, spatial navigation (fMRI of taxicab drivers increased activity), speech comprehension.

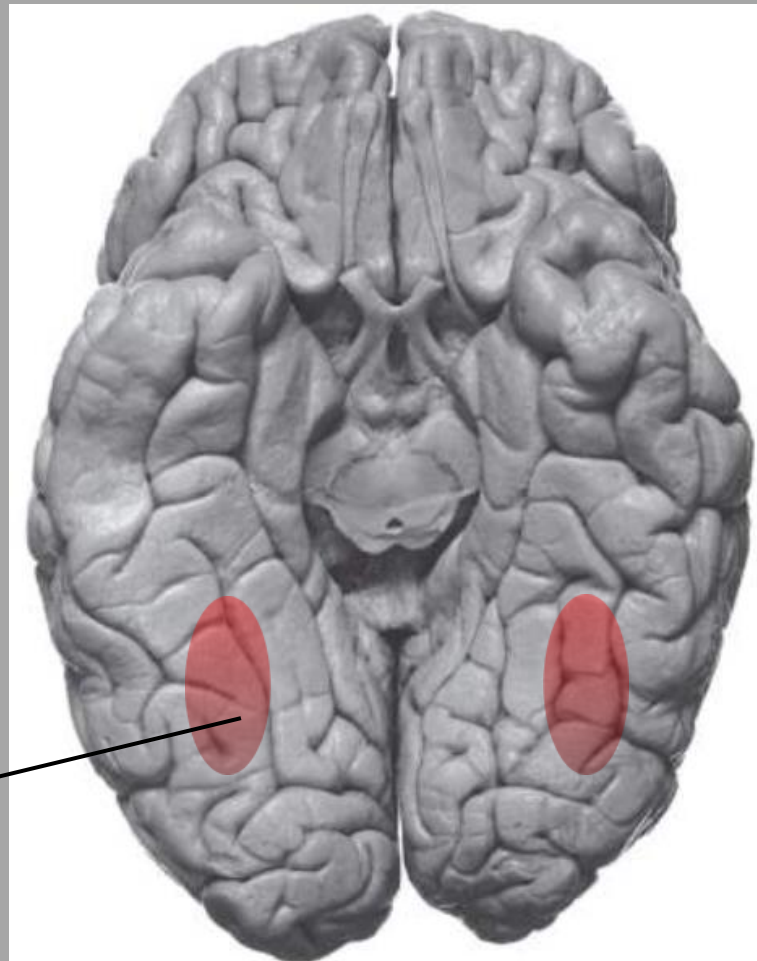
- Primary auditory cortex → damage leads to auditory deficits
- Wernicke's aphasia: in the inability to comprehend spoken language and often confabulation (also called 'receptive aphasia')
 - Area helps us listen to speech and then respond propel with fluent language .





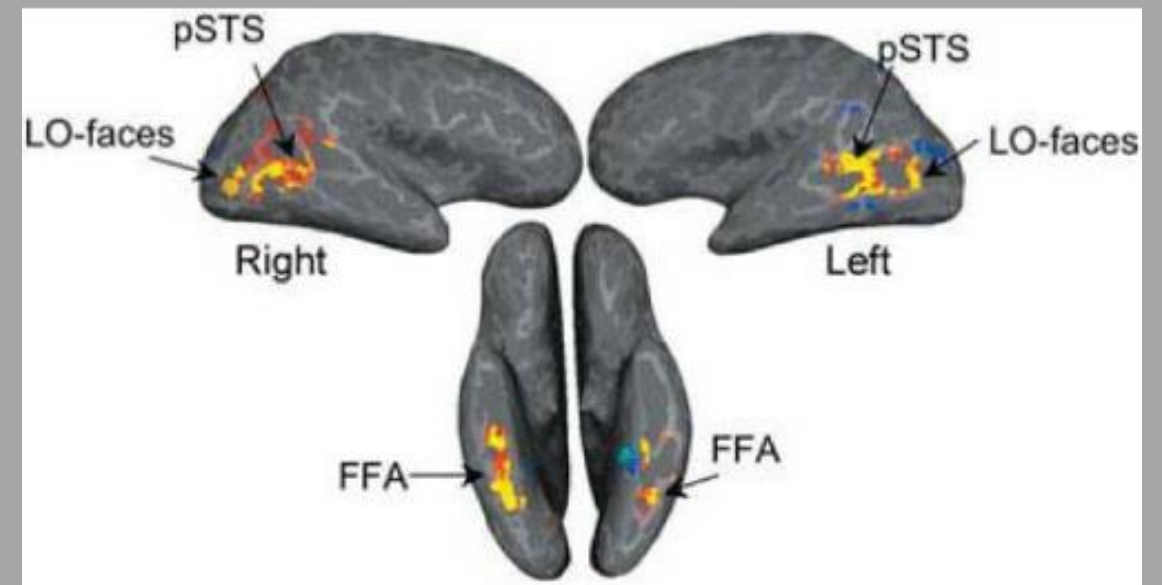
Temporal lobe:

Audition (future lecture) & memory , higher order visual processing, spatial navigation.



Temporal lobe areas that show increased activation following presentation of faces:

1. Fusiform Gyrus/Face Area (FFA),
2. Superior Temporal Sulcus (STS)
3. Lateral occipital area



FFA damage



Prosopagnosia: Inability to recognize faces after damage to inferotemporal lobe (FFA). Blur of face when pt is asked. Still recognize voice of person. Different neural pathway involved with that

Behavioral/Systems/Cognitive

Electrical Stimulation of Human Fusiform Face-Selective Regions Distorts Face Perception

Josef Parvizi,^{1,2} Corentin Jacques,^{2,3,4} Brett L. Foster,^{1,2} Nathan Withoft,^{2,4} Vinitha Rangarajan,^{1,2} Kevin S. Weiner,^{2,4} and Kalanit Grill-Spector^{2,4}

- Electrical implants placed into fusiform face area
- Stimulation perturbed FFA neuron physiology and distorted pts perception of the clinician's face
- FFA increased blood flow on fMRI when showing a person pictures of faces



Video



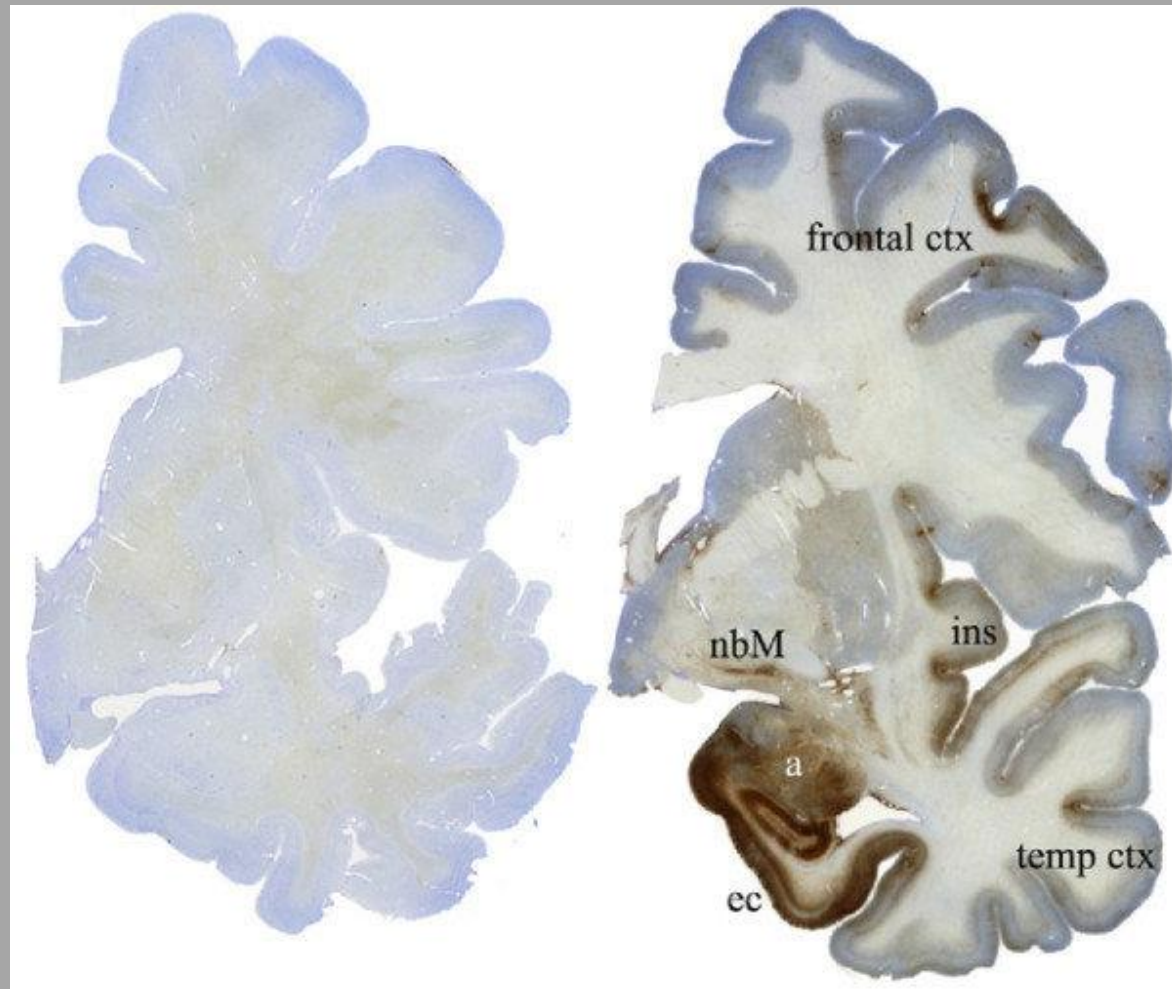
Medial Temporal Lobe

MTL vulnerable to injury after head trauma

Tau staining → neuronal injury marker

Normal

Pro football player

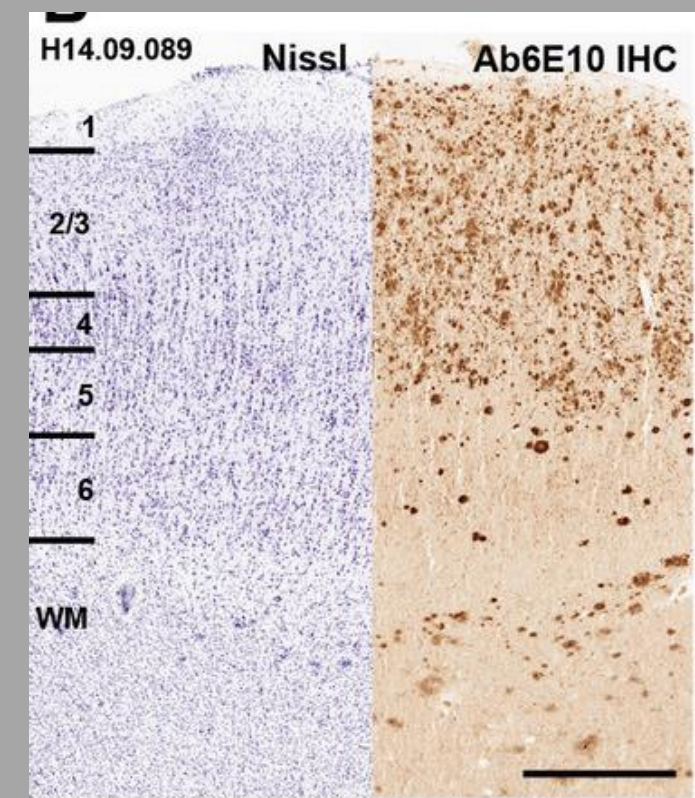
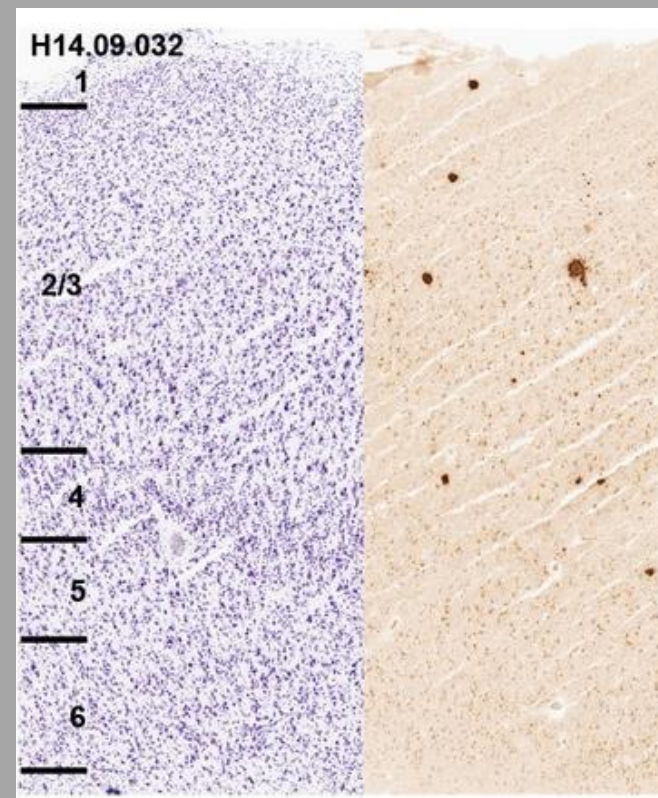


MTL vulnerable to injury in Alzheimer's disease

- Whether Tau accumulation is the cause of memory loss or some event preceding tau creation is essentially unknown at this point
- Damage/neuronal pathology correlated with loss of memory function

Normal

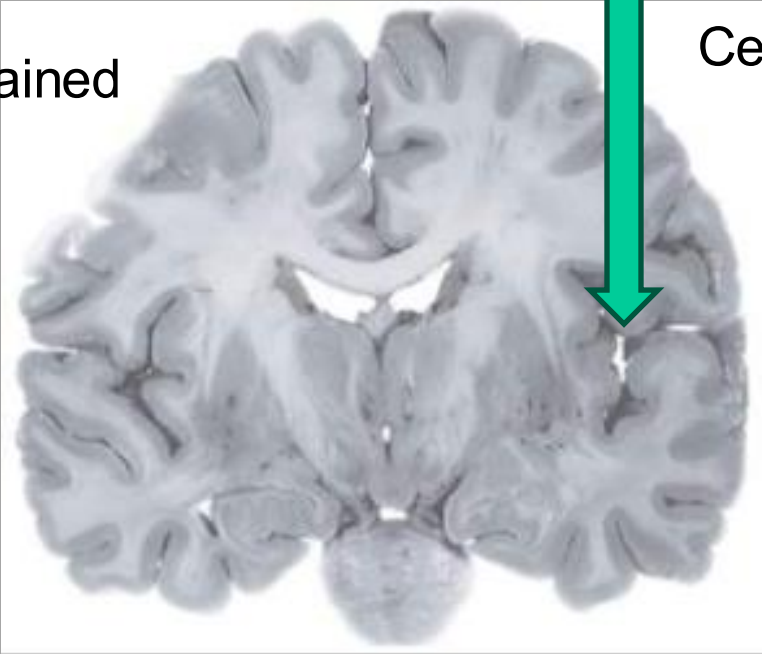
Patient w/ AD



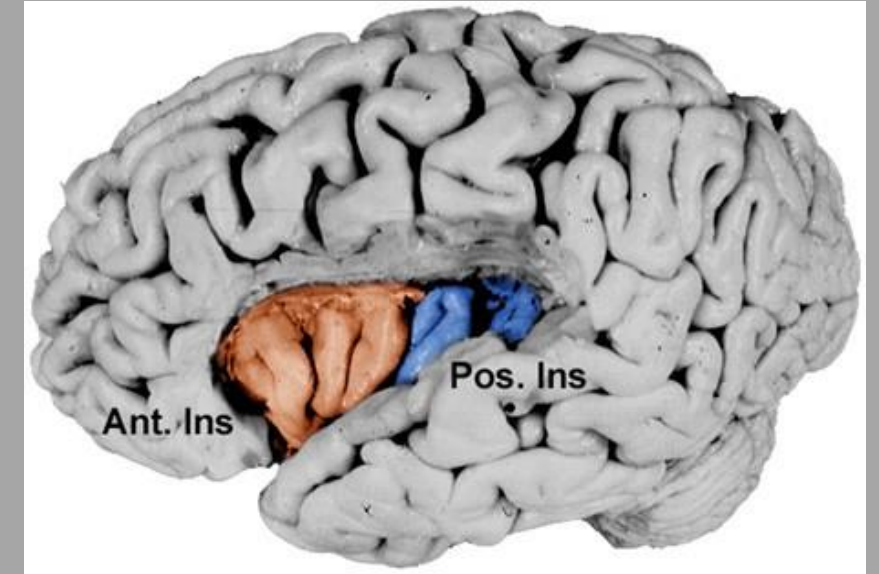
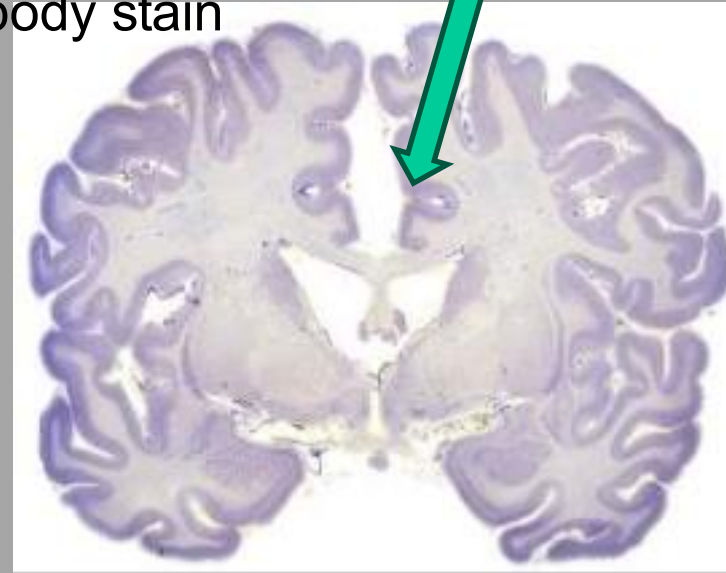
Neuropathological and transcriptomic characteristics of the aged brain. Miller JA, et al. Elife. 2017. PMID: 29120328

Insular Cortex and Cingulate Cortex

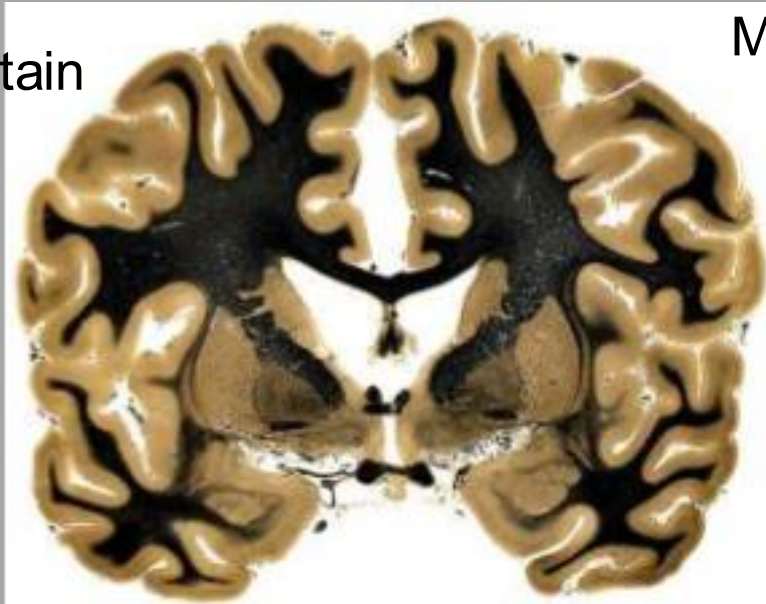
Unstained



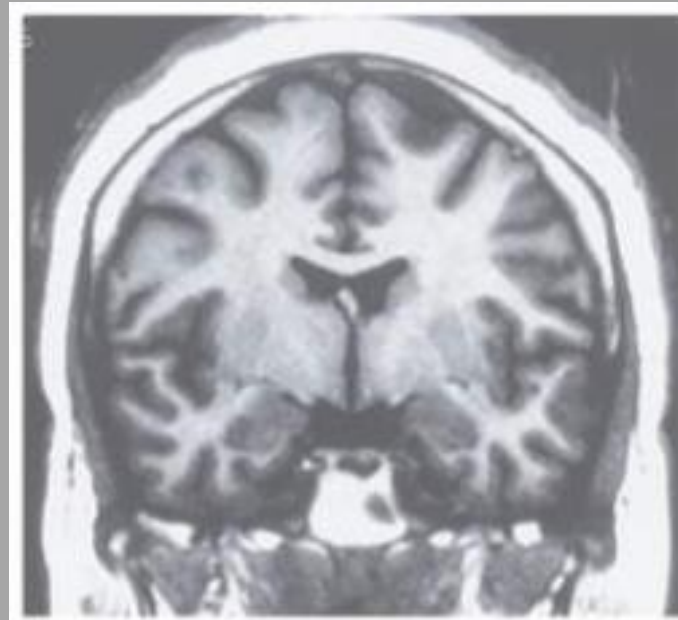
Cell body stain



Axonal stain



MRI



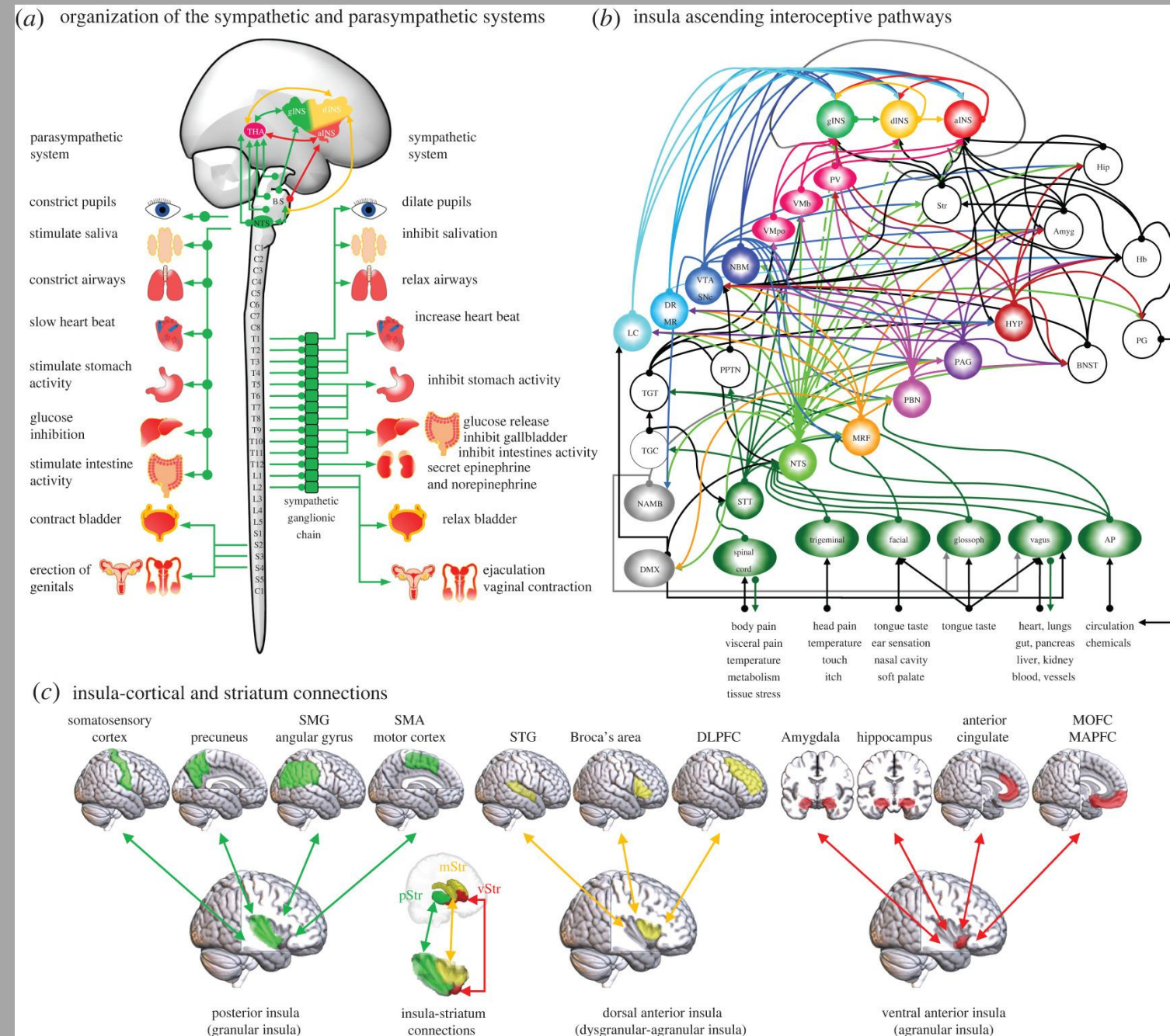
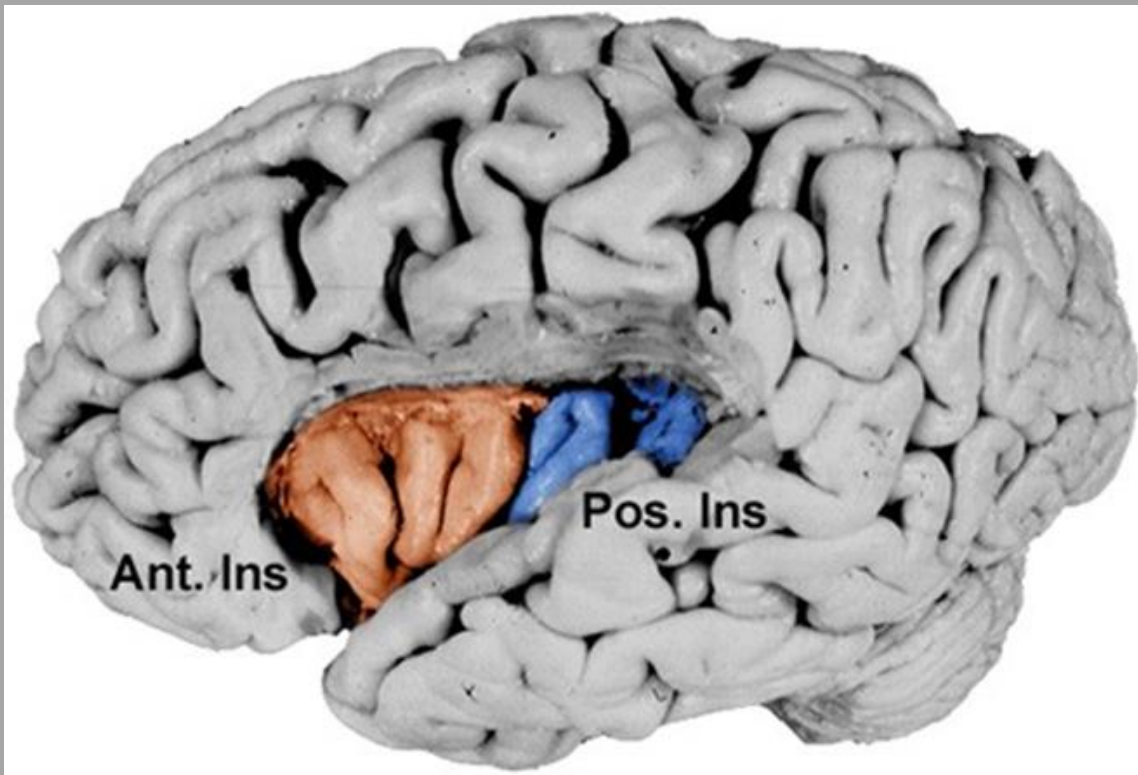
- Poorly understood areas compared to primary sensory areas

Insular cortex- diverse functions including: Gustation, olfaction, vestibular, pleasure, reward & addiction, visceral & autonomic functions. Receives input related to these areas. A hub for interoception and translation interoceptive inputs into conscious feelings. Damage to insular lobe can cause different clinical pictures.

Connectivity of the Insular cortex

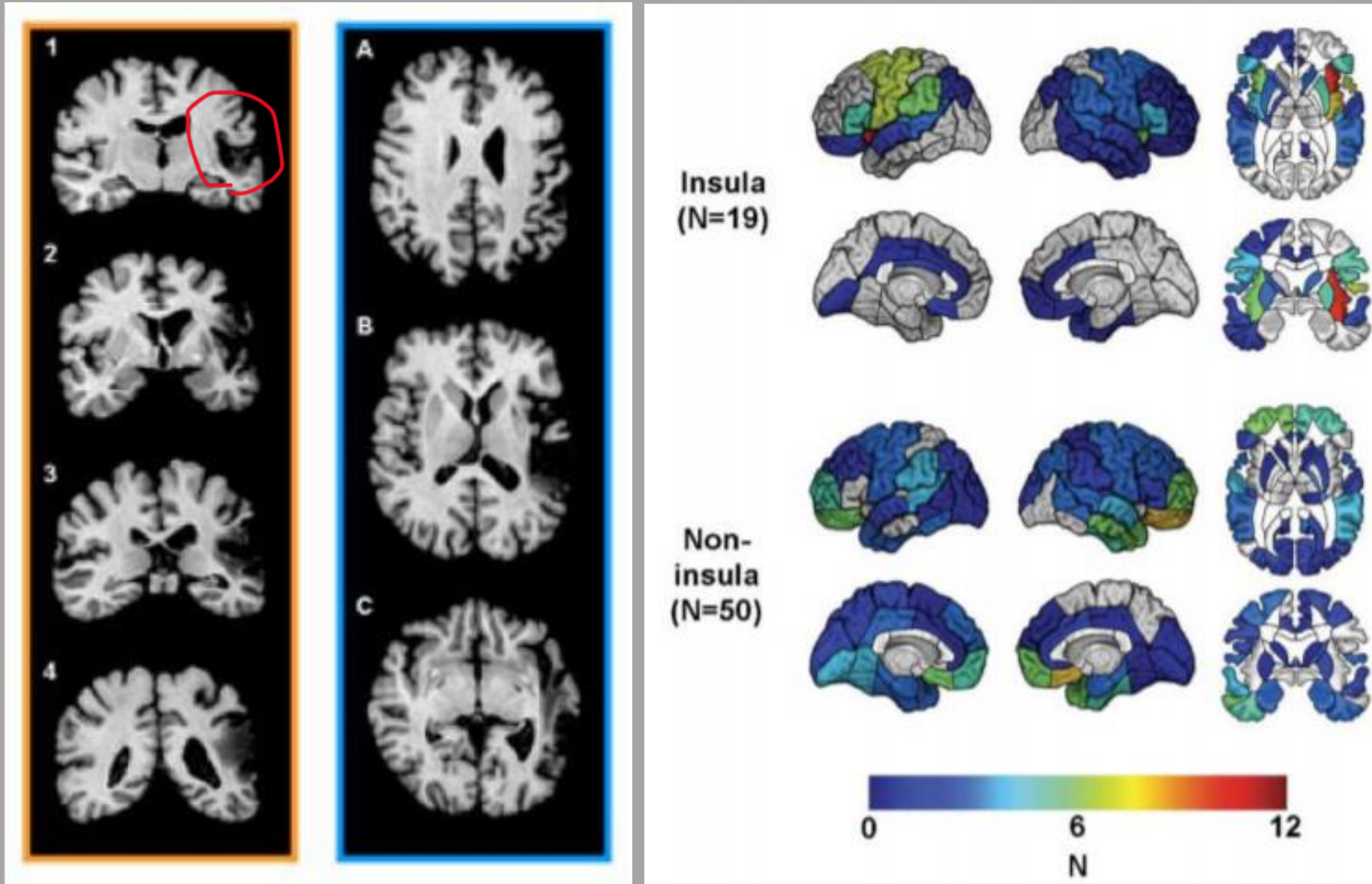
Fermin AS et al. 2022

- Insular cortex is activated in tasting rancid food as well as stimuli that are deemed to be morally disgusting (violence) via fMRI. Robert M. Sapolsky. (2023). *Determined: A Science of Life without Free Will*. Penguin Publishing Group.



Damage to the Insula Disrupts Addiction to Cigarette Smoking

Nasir H. Naqvi¹, David Rudrauf^{1,2}, Hanna Damasio^{3,4}, and Antoine Bechara^{1,3,4,*}



- Insula is necessary for the ability to consciously appreciate and derive pleasure
- Necessary for the sensation of urges – like the urge to smoke seen in smokers.
- Possibly area to target with novel drugs or TMS in the future.

Electrical stimulation of the insula elicits taste sensation

Jennifer Yih et al *Social Cognitive and Affective Neuroscience*, Volume 14, Issue 4, April 2019, Pages 339–351, <https://doi.org/10.1093/scan/nsz015>



Pt undergoing
neurosurgery. Insula is
being stimulated

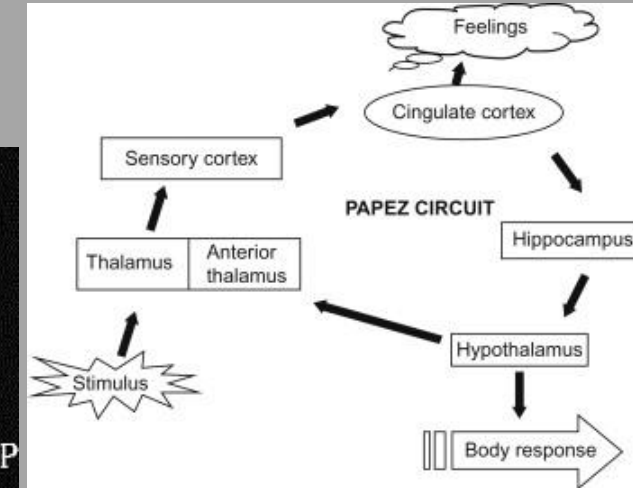
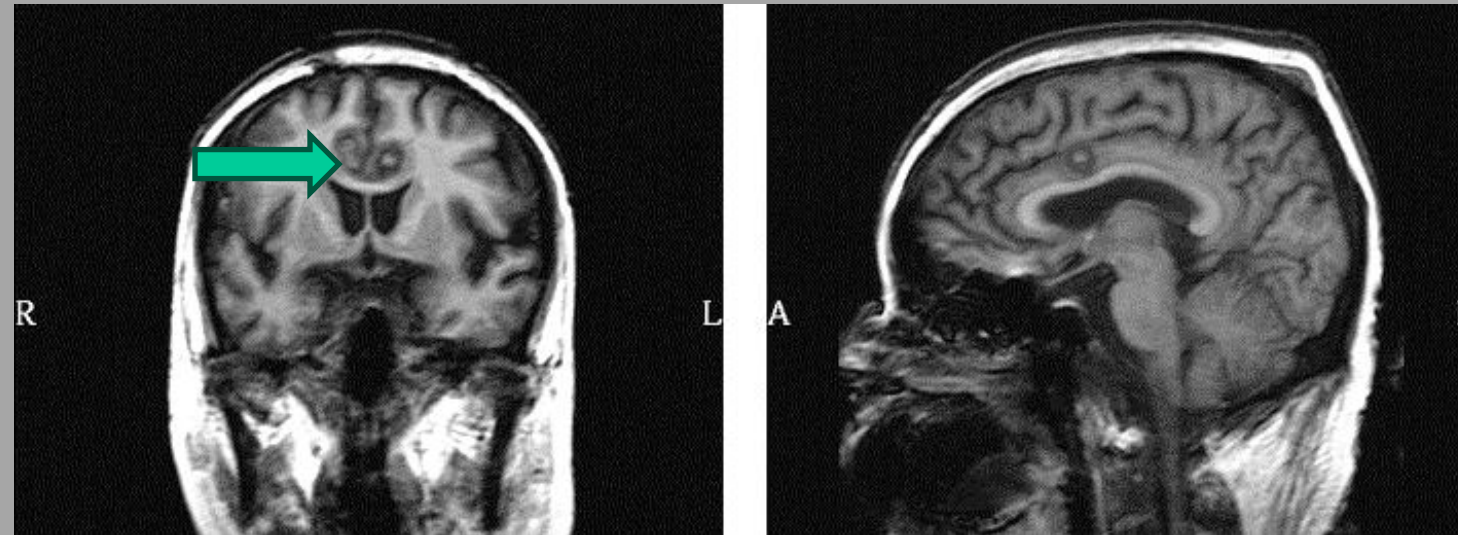
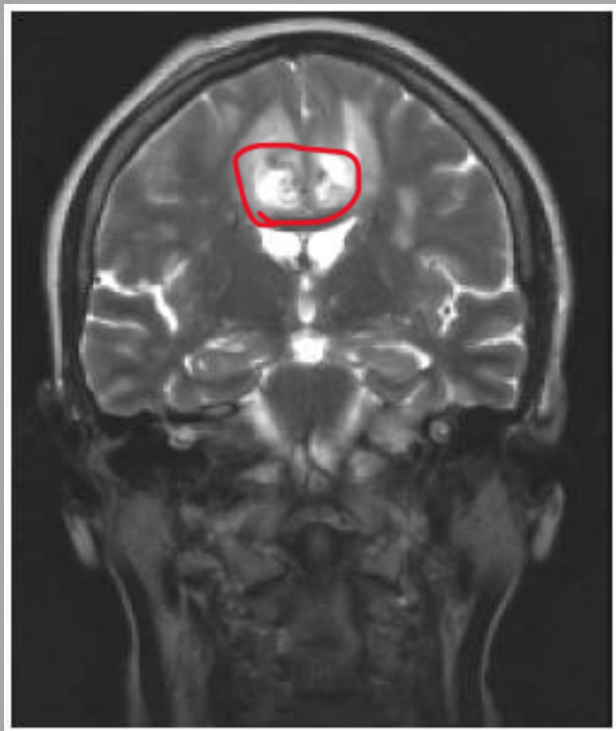
Subject 15 was an especially compelling case study as she reported a chocolate taste at lower intensity, which became a distasteful 'bad' chocolate taste with increased dose and duration of stimulation to the same electrode pairing. In one instance, subject 15 even made a disgusted expression with her face, which had been neutral in prior stimulations ([Video 1](#)).

Cingulate cortex: cognition, emotion, memory (Papez circuit), 4F's

- Feeding, fighting, flighting, fornicating

Bilateral anterior
Cingulotomy for drug-
resistant depression or OCD

Bilateral anterior Cingulotomy
for the treatment of chronic pain



This patient suffered from metastatic bone pain which was completely eliminated after surgery and the effect remained for 20 months. (Yen CP et al. 2005)

1 wk after surgery; Sam-Eljamel 2008

Surgery is safe and can be effective depending on numerous factors...

Cingulotomy also considered for Postherpetic Neuralgia & Trigeminal Neuralgia

Cingulotomy performed at Columbia U/ NY Presbyterian in NYC

www.neurosurgery.columbia.edu/patient-care/treatments/cingulotomy

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Cingulotomy

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When is Cingulotomy performed?

Appropriate candidates for cingulotomy are individuals for whom traditional medical and/or surgical therapies have been unsuccessful. Cingulotomy is a non-reversible procedure and is therefore reserved for those patients or whom all other options have been exhausted.

In chronic pain patients, the greatest benefit is usually experienced by those whose pain is related to stroke, cancer, or spinal cord injury. Generally, pain must have been present for a period of at least six months and multiple conventional therapies must have already been tried.

Individuals with OCD must have medically refractory, severe OCD. A multidisciplinary team of specialists from psychiatry, psychology, neurology, and neurosurgery reviews cases to determine whether surgery may be appropriate.

How should I prepare for Cingulotomy?

Modern cingulotomy is most often performed using thermal energy or specialized laser systems. No large open incision is required. and many patients are able to go home the following day.

Research report

Mirth and laughter elicited by electrical stimulation of the human anterior cingulate cortex

Fausto Caruana ^{a, b} ✉, Pietro Avanzini ^{b, c}, Francesca Gozzo ^d, Stefano Francione ^d, Francesco Cardinale ^d, Giacomo Rizzolatti ^{a, b}



Patient	Hem.	Description
Patient 1	Right	Burst of laugh
Patient 2	Right	Burst of laugh
Patient 3	Left	Burst of laugh
Patient 4	Left	Burst of laugh
Patient 5	Left	Burst of laugh
Patient 6	Right	Laugh w/out mirth
Patient 7	Right	Laugh w/out mirth
Patient 8	Right	Laugh w/out mirth
Patient 9	Left	Laugh w/out mirth
Patient 10a	Right	Laugh w/out mirth
Patient 10b	Right	Laugh w/out mirth

Summary slide

- The entire neocortex has a characteristic cytoarchitecture based on lamina of radially-oriented pyramidal cells with diverse projections.
- Different cortical areas produce sensory, motor, or cognitive functions specific to particular modalities. These functional areas are created by the synaptic connections made within and across the neocortex as well as from synaptic input from subcortical brain regions. I.e thalamus
- Injury to a particular neocortical brain area will result in specific change in behavior or cognitive function which can be identified by a physician.

Question

- A 55-year-old man is brought into the family medicine clinic because of recent change in behavior over the last 2 years. Once a sweet and reserved man the patient is now yelling obscene comments at his coworkers on the job. His wife also reports that he is gotten much ruder, and less empathetic to her needs over the last 2 years as well. He also has spent \$40,000 in the last month gambling. What is the most likely diagnosis and gene associated with the diagnosis
- A) PARK2- Parkinson's
- B) HTT- Huntington's
- C) APOE- Alzheimer's Disease
- D) SOD1– Amyotrophic lateral Sclerosis
- E) MAPT Frontotemporal Dementia

Answer

- E) MAPT Frontotemporal Dementia

Question

- A 38-year-old man gets hit in the head by a rock that was thrown at him. It is noticed that the man cannot voluntarily move his eyes from side to side. What brain area is most likely affected?
- A) Temporal lobe (hippocampus)
- B) Fusiform gyrus
- C) Broca's Area
- D) Wernicke's Area
- E) Frontal Area 8

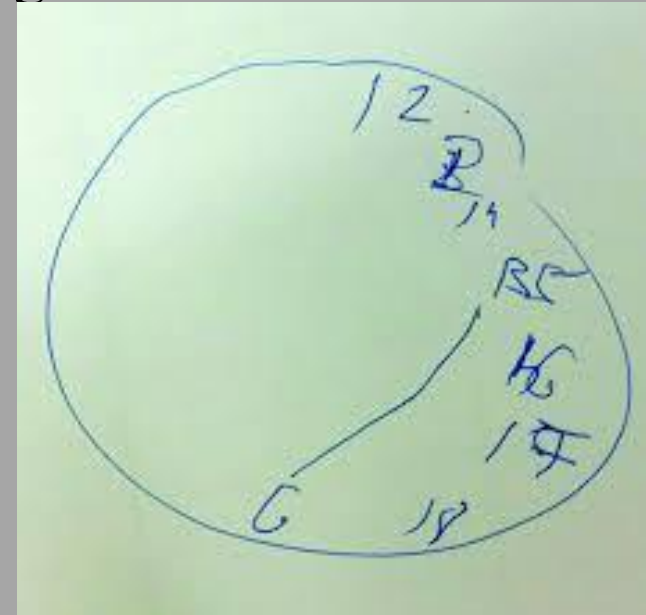
Answer

- E) Frontal Area 8

Question

- A pt suffers an ischemic stroke of the Right MCA. About a week after the stroke while the pt is recovering in the hospital, a family member noticed the pt only eating food in the right side of his plate. when asked to draw a clock he does this. What's the most likely diagnosis?

- A) Prosopagnosia
- B) Broca's aphasia
- C) Angular gyrus lesion
- D) Gerstmann's syndrome
- E) Hemi-spatial neglect



Answer

- E) Hemi-spatial neglect

Question

- A 27 year old patient is undergoing neurosurgery and the neurosurgeon is placing electrodes into the brain of a patient. In order to map out a pathway to dissect a tumor properly, the surgeon is stimulating a certain region of the brain in order to map what the function of the specific area is. Upon heavy electrical stimulation, the patient describes tasting rancid butter and is clearly in disgust. What area was most likely stimulated area?
- A) Fusiform face area
 - B) Insular cortex
 - C) Broca's area
 - D) Frontal lobe area 8
 - E) Angular gyrus

Answer

- B) Insular cortex

Question

- A patient suffers an ischemic stroke in a branch of the left MCA. Upon recovering in the hospital, a family member has noticed that the patient is having trouble calculating in how many hours his daughter will visit him. A nurse on staff also notes that the patient is now confusing his left side from his right side and that he keeps stating his middle fingers are his pinkie fingers. What brain region is most likely damaged from this stroke?
- A) Non dominant parietal lobe (Right)
 - B) Temporal lobe
 - C) Occipital lobe
 - D) Insular cortex
 - E) Left angular gyrus (left inferior parietal lobe)

Answer

- E) Left angular gyrus (left inferior parietal lobe)

Feedback is welcomed

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>

